Management Explanation Report

Generated directly from main.ipynb

Notebook source: main.ipynb

Charts extracted: 34

Generated on: 2026-03-21

This version is built from notebook outputs only.
It does not re-read the CSV during PDF generation.

Flow

1. Data meaning (business terms)
2. Graph-by-graph explanation
3. Example output lines from notebook

Data First: What Each Field Means

Volume in this report means: Total Qty (shipment units) aggregated over a group (year, quarter, country, department, etc.).

- `Invoice Date` = shipment/invoice date
- `Financial Year` = reporting year bucket (FY)
- `Quarter` = fiscal quarter (Q1 to Q4)
- `Final Country` = destination market/country
- `Department` = internal commercial/operational owner
- `Strength In MG` = dosage strength (e.g., 500 mg, 1000 mg)
- `Market` = channel type (Tender, Private, Other)
- `Pharmacopoeia` = regulatory standard (EP/USP/IP)
- `Total Qty` = units shipped (this is the core volume measure)
- `API Requirement in KG` = API needed for produced/shipped quantity
- `Wastage` = process factor used with API/quantity calculations

How to explain each graph to management:

1) Start with what the graph measures.
2) Point to trend or concentration.
3) Give one concrete example number.
4) End with business action.

1) Year-over-Year (YoY) Growth Analysis

Notebook cell: 3

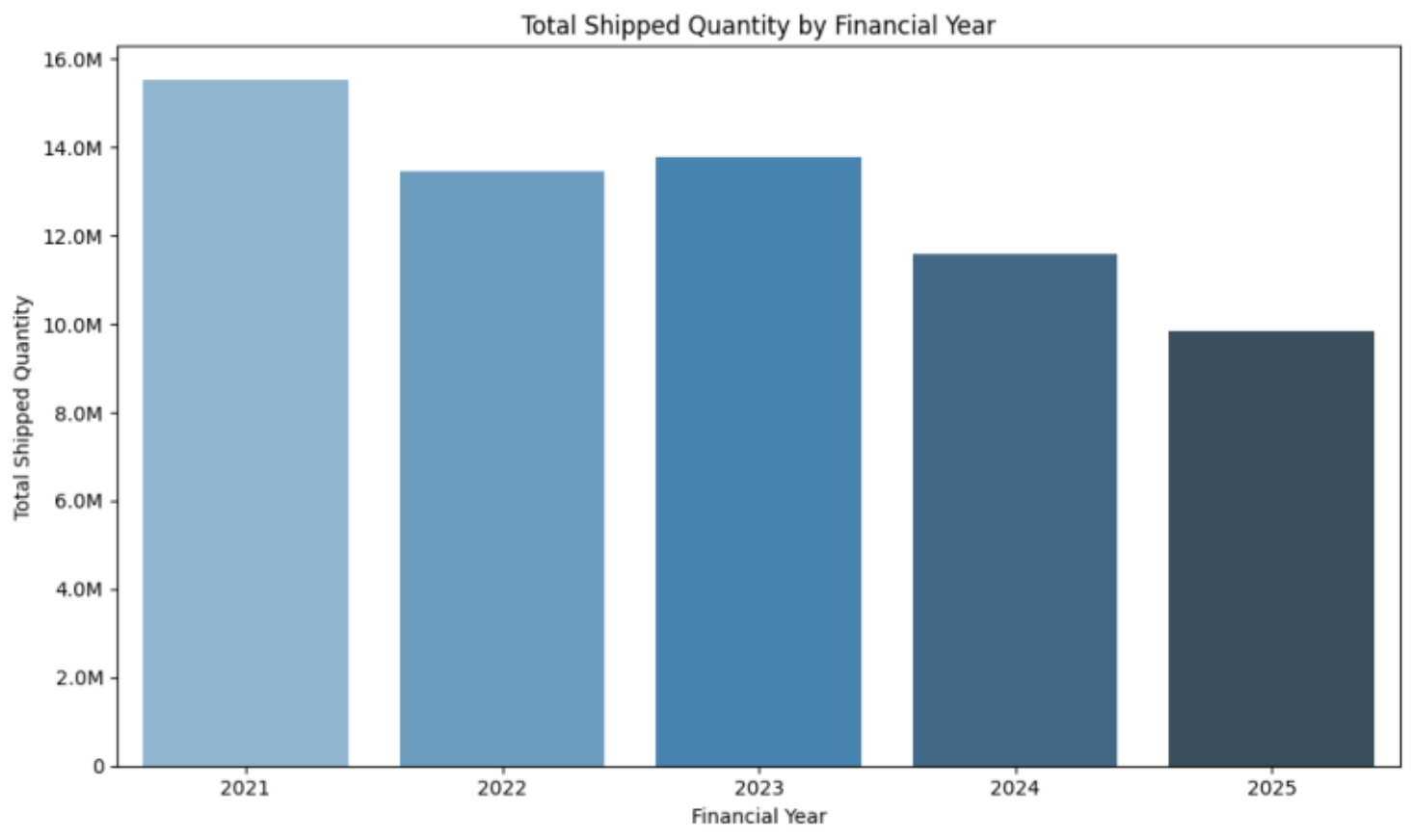
How To Explain This Graph

What this shows: Annual volume trend by financial year using Total Qty.

How to read it: Compare each year bar to explain growth/decline and overall demand direction.

Example From Notebook Output

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.
sns.barplot(x=yoy_growth.index, y=yoy_growth.values, palette='Blues_d')



2) Quarterly Seasonality Pattern

Notebook cell: 4

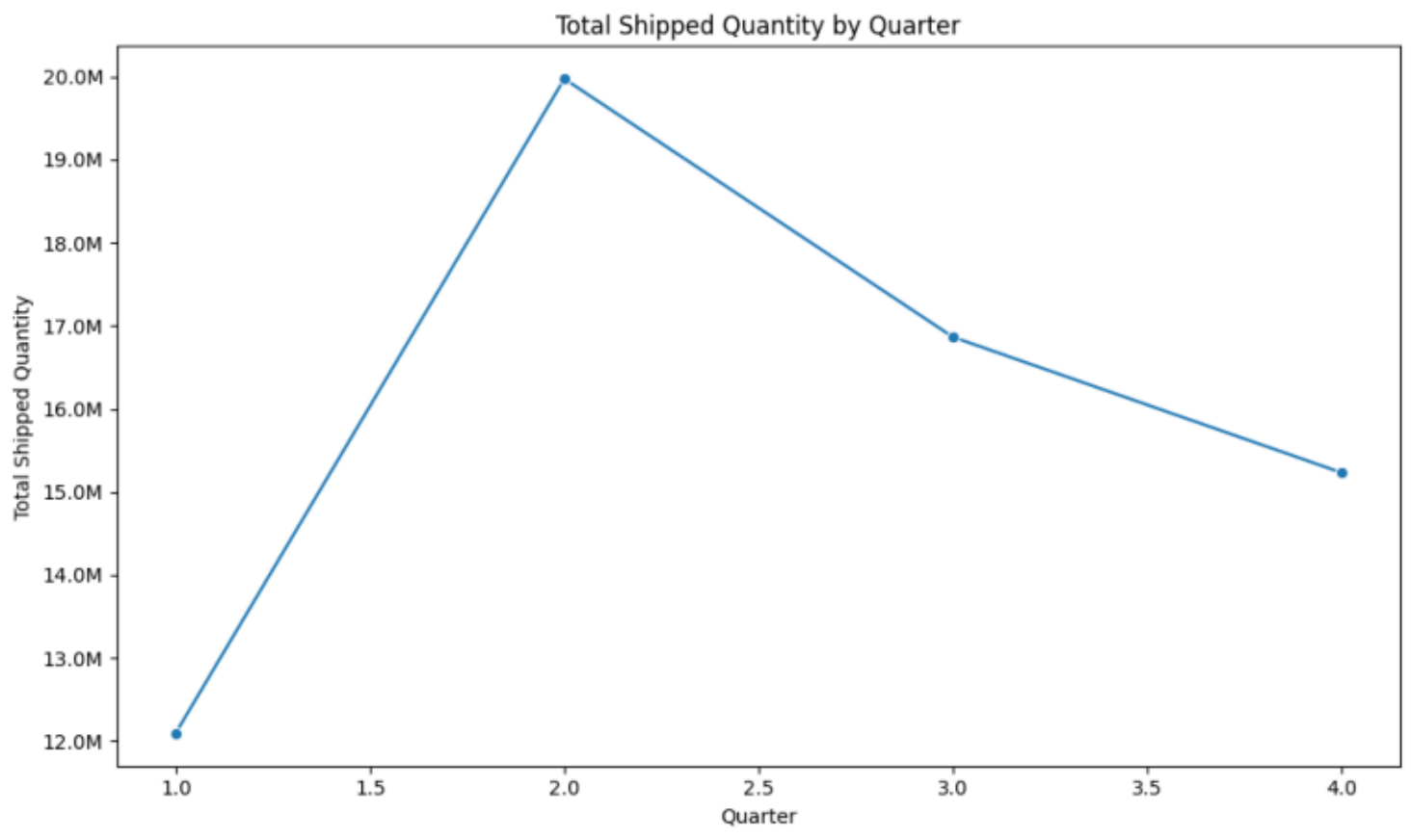
How To Explain This Graph

What this shows: Quarter-level seasonality pattern for total shipped units.

How to read it: Highlight recurring peak quarters and weak quarters for inventory/dispatch planning.

Example From Notebook Output

No textual output for this chart in notebook (visual-only output).



3) Strength Distribution Mix

Notebook cell: 5

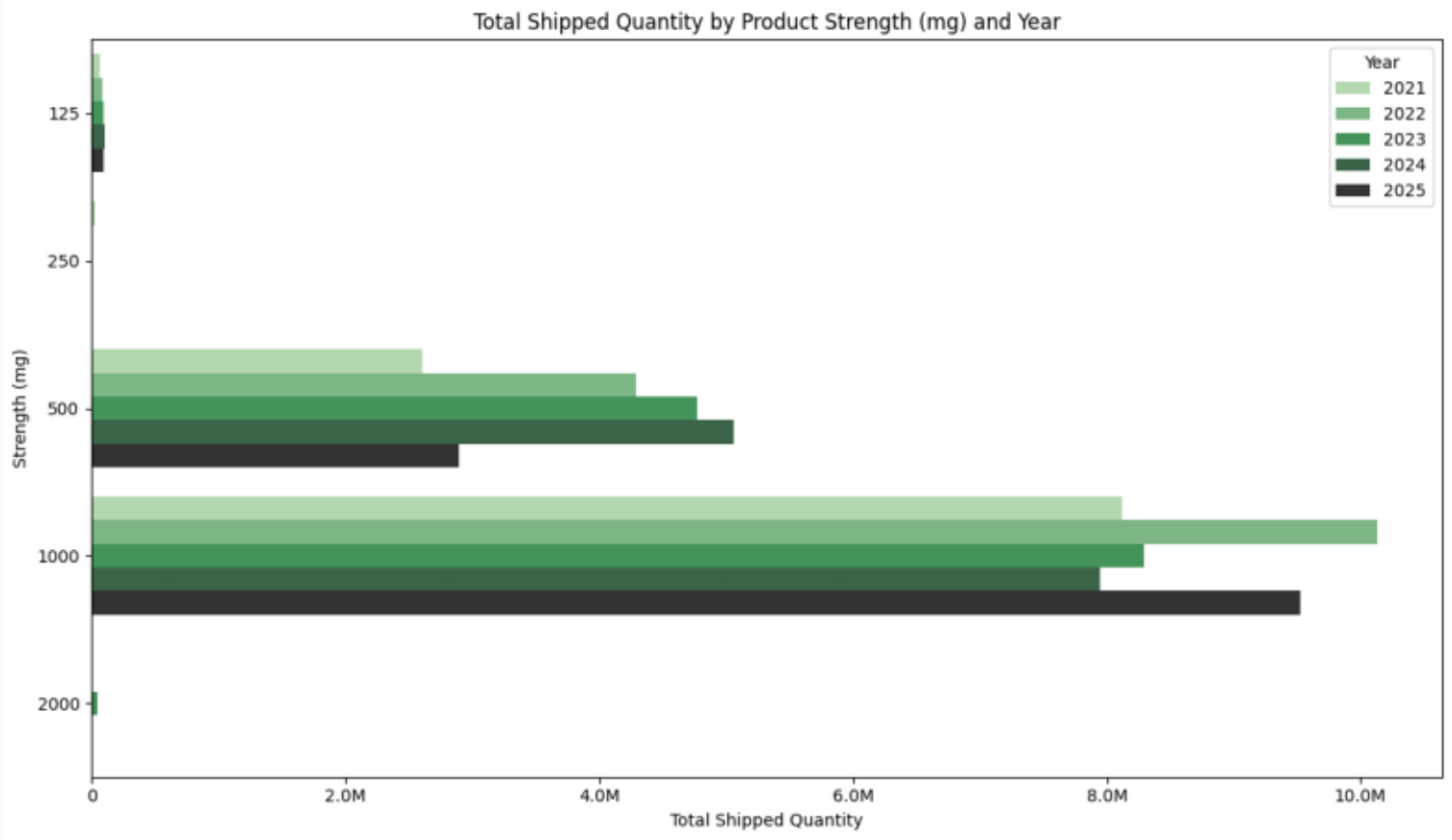
How To Explain This Graph

What this shows: Strength-wise volume distribution across years (125/250/500/1000/2000 mg).

How to read it: Explain where demand is concentrated and whether strength preference is shifting.

Example From Notebook Output

No textual output for this chart in notebook (visual-only output).



4) Geographic Concentration Risk

Notebook cell: 6

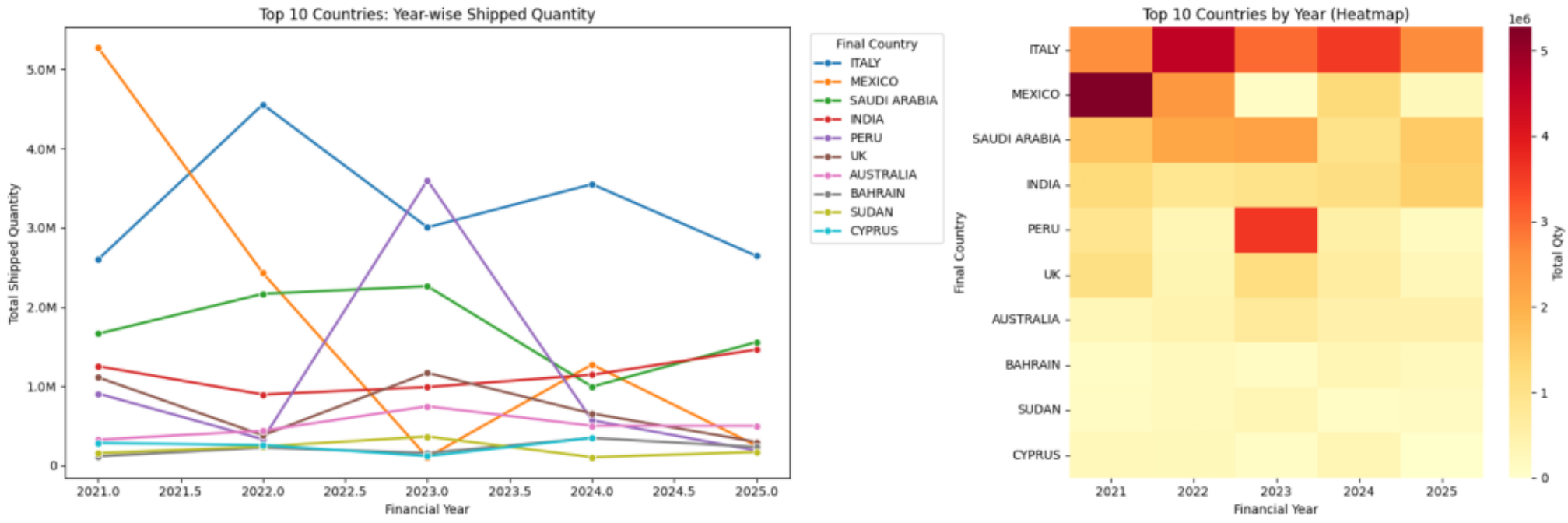
How To Explain This Graph

What this shows: Geographic concentration view for top countries over time.

How to read it: Use line trends + heatmap intensity to show concentration risk and country momentum.

Example From Notebook Output

No textual output for this chart in notebook (visual-only output).



5) Departmental Volume Leaderboard

Notebook cell: 7

How To Explain This Graph

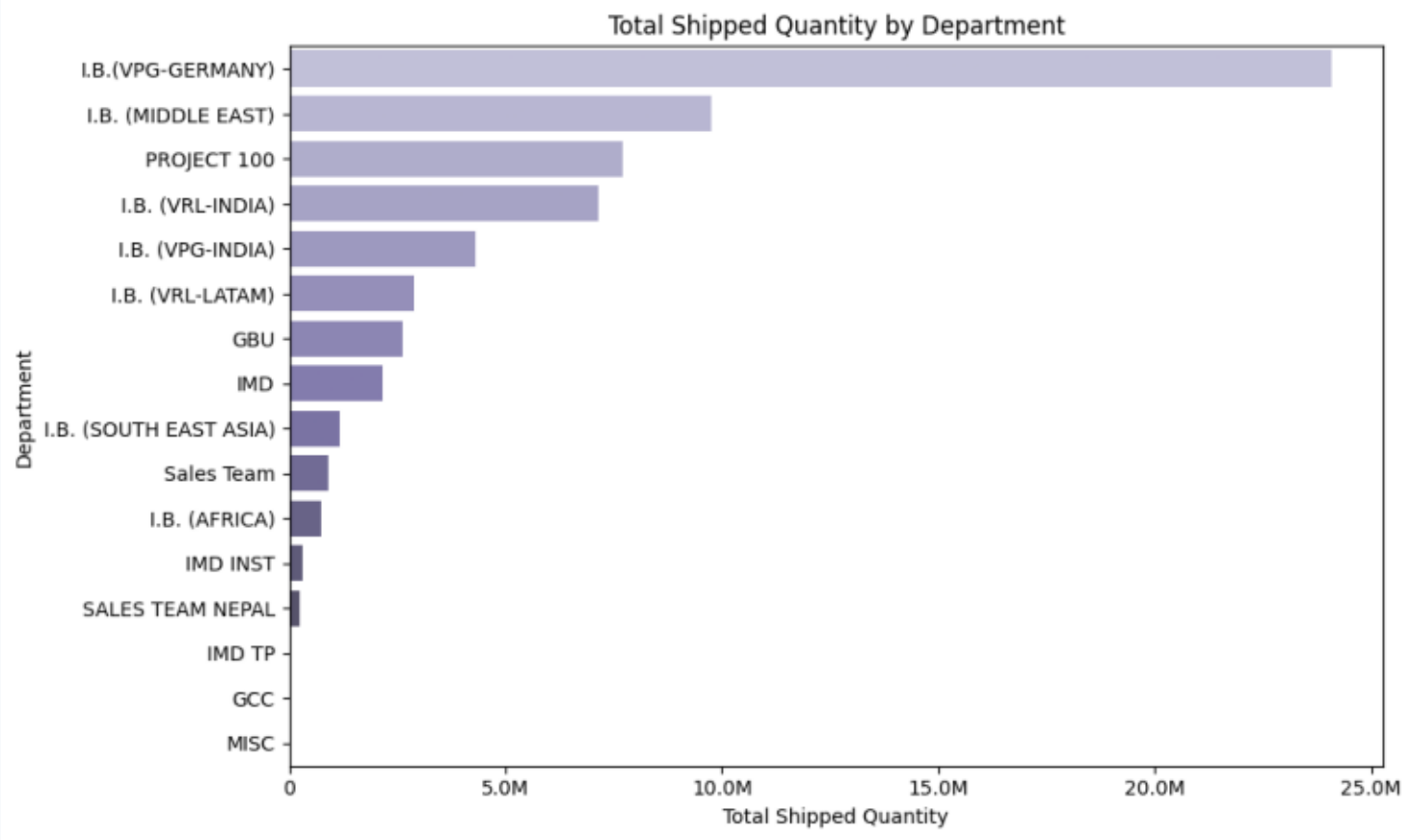
What this shows: Department leaderboard by shipped volume.

How to read it: Explain top contributors and discuss resource allocation around high-volume teams.

Example From Notebook Output

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the same effect.

```
sns.barplot(x=dept_performance.values,
y=dept_performance.index, palette='Purples_d')
```



6) Market Share Breakdown (Tender vs Private vs Other)

Notebook cell: 8

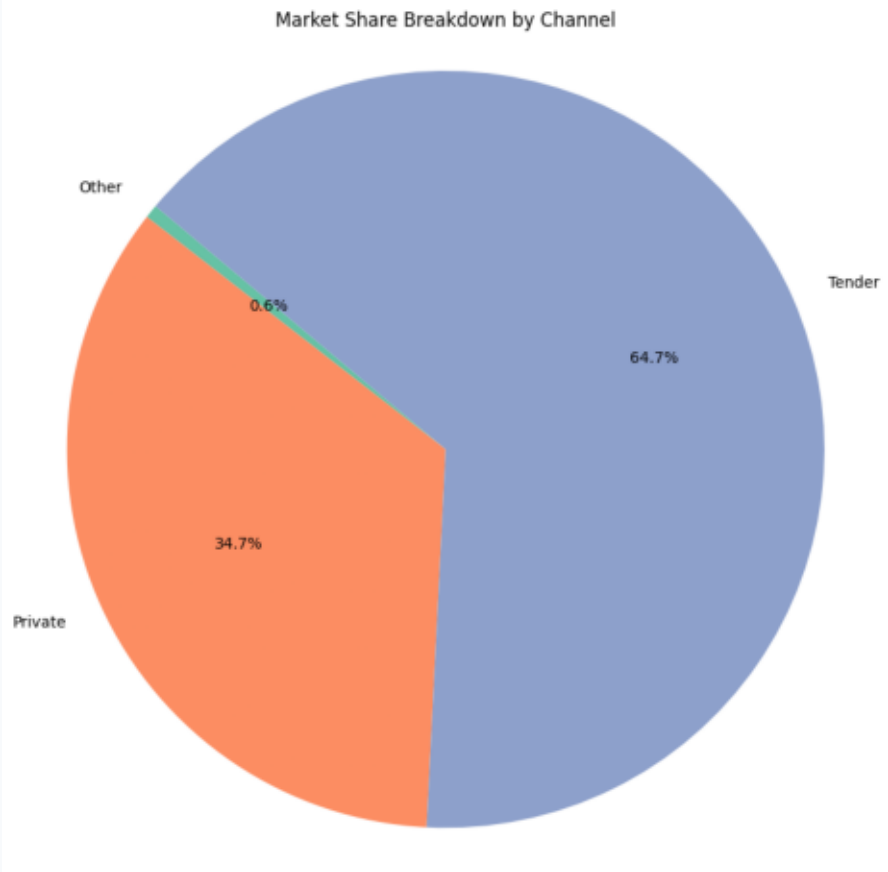
How To Explain This Graph

What this shows: Overall market channel split: Tender vs Private vs Other.

How to read it: Use percentage shares to explain channel dependence and diversification scope.

Example From Notebook Output

No textual output for this chart in notebook (visual-only output).



7) Pharmacopoeia Compliance Overview

Notebook cell: 9

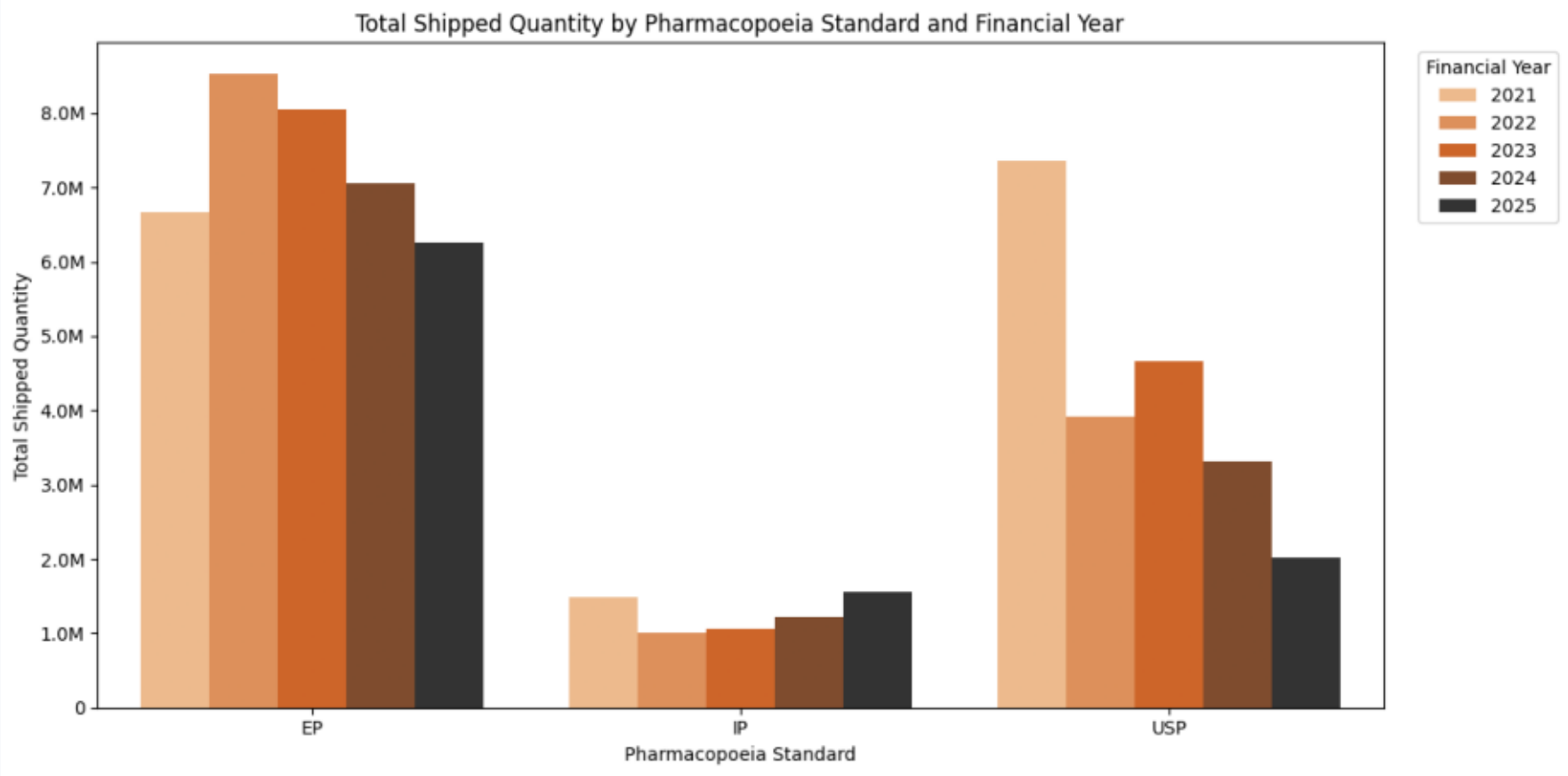
How To Explain This Graph

What this shows: Volume distribution by pharmacopoeia standard and year.

How to read it: Show compliance mix and where quality/regulatory load is highest.

Example From Notebook Output

No textual output for this chart in notebook (visual-only output).



8) Strategic Project Lifecycle Analysis (PROJECT 100)

Notebook cell: 13

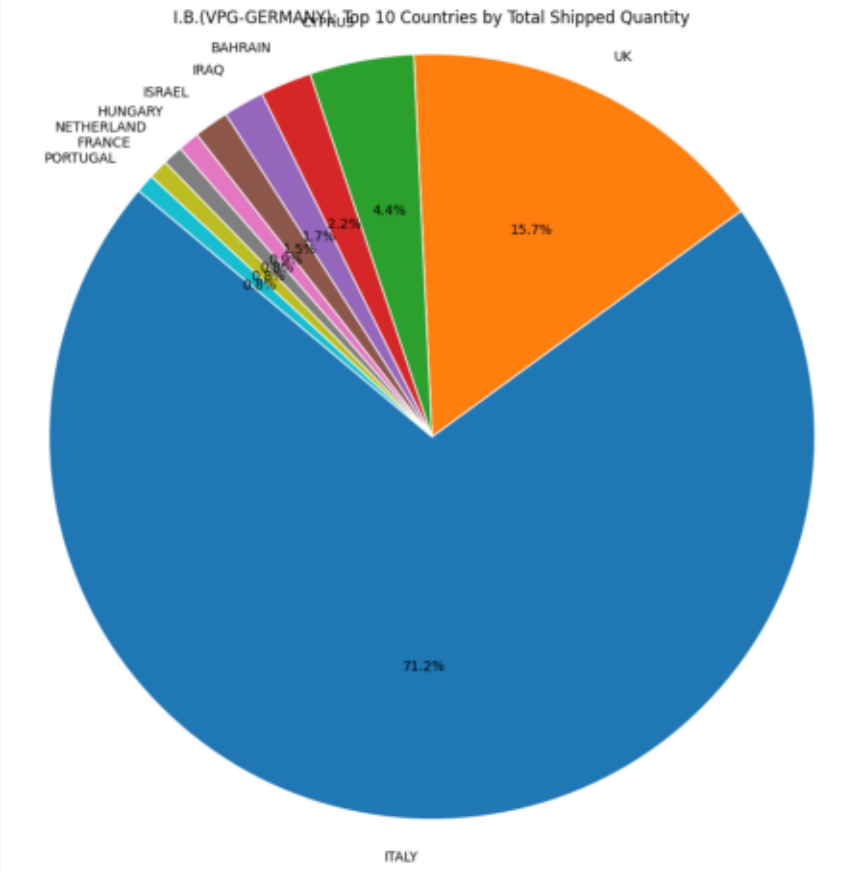
How To Explain This Graph

What this shows: Country split for I.B.(VPG-GERMANY) strategic/project volume.

How to read it: Use this to explain where that department's portfolio is concentrated geographically.

Example From Notebook Output

No textual output for this chart in notebook (visual-only output).



12) Market Transition Trends for India

Notebook cell: 17

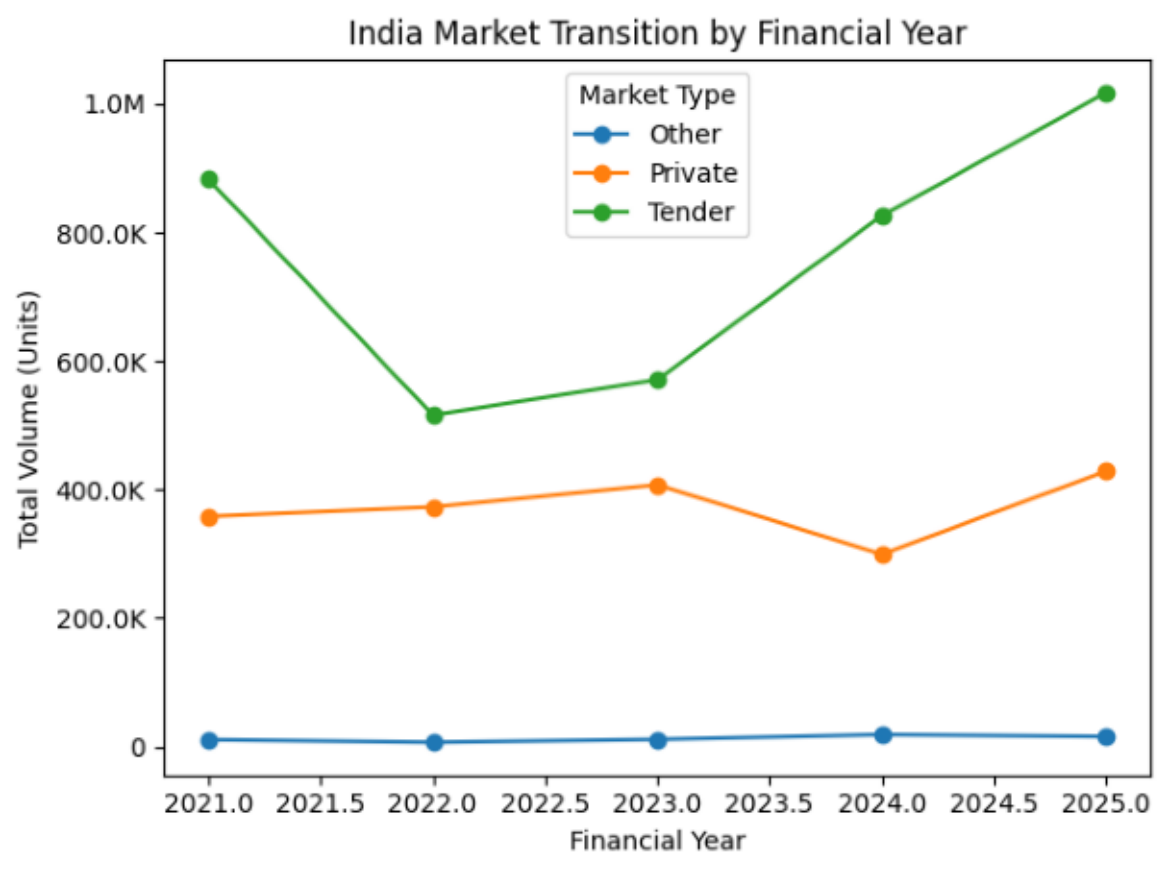
How To Explain This Graph

What this shows: India channel transition trend by year.

How to read it: Discuss whether India is moving between Tender and Private and the commercial implication.

Example From Notebook Output

No textual output for this chart in notebook (visual-only output).



14) Departmental Market Share Internal Competition (Year-wise)

Notebook cell: 19

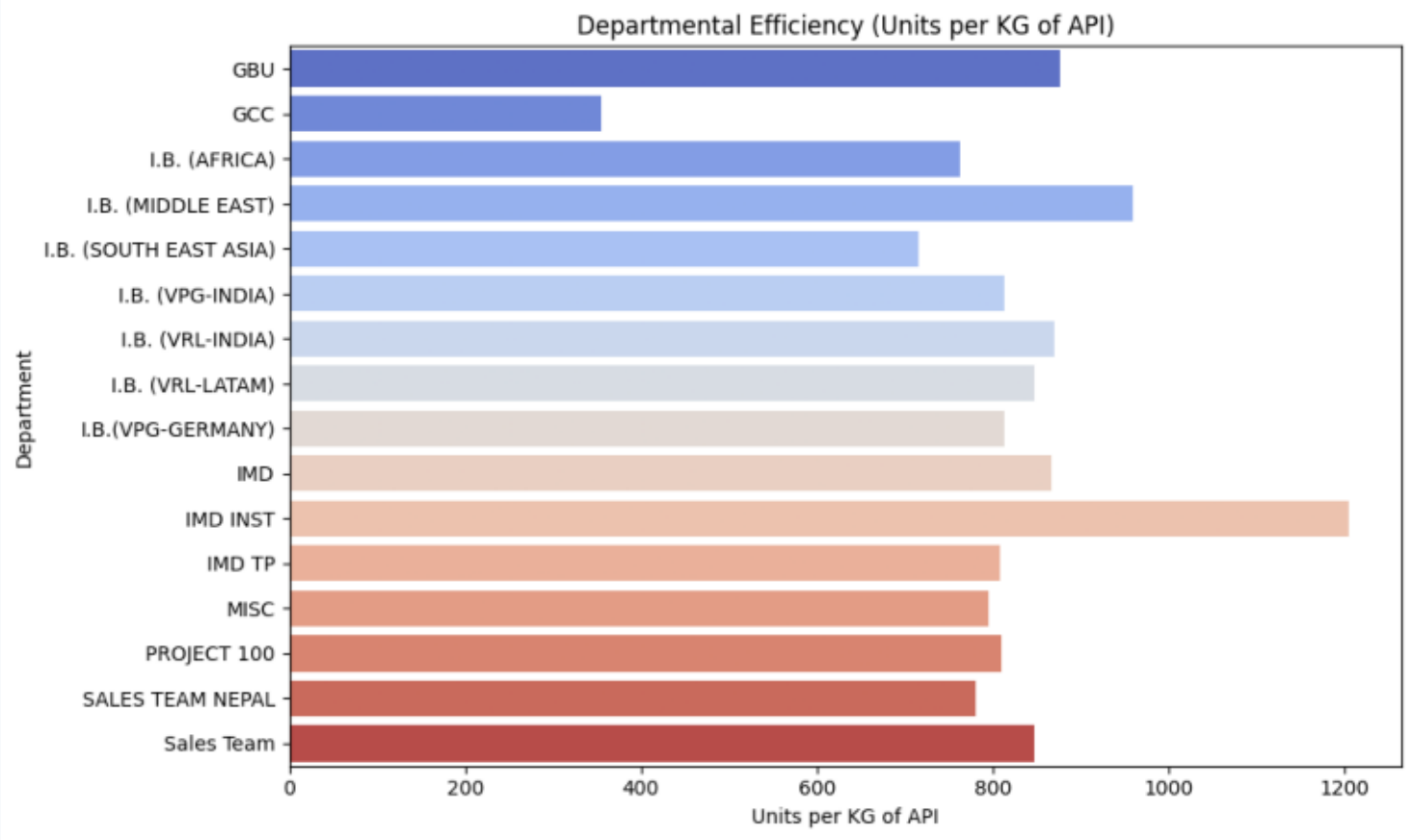
How To Explain This Graph

What this shows: Department efficiency comparison (units delivered per kg API).

How to read it: Higher values imply better API conversion efficiency; compare leaders vs laggards.

Example From Notebook Output

Departmental Efficiency (Units per KG of API):
Total Qty API Requirement in KG API_Efficiency
Department
IMD INST 320210
265.90485 1204.227753
I.B. (MIDDLE EAST) 9763616
10181.87720 958.921013
GBU 2636356
3006.00894 877.028662



19) Lead Time and Re-order Cycles (for recurring countries)

Notebook cell: 24

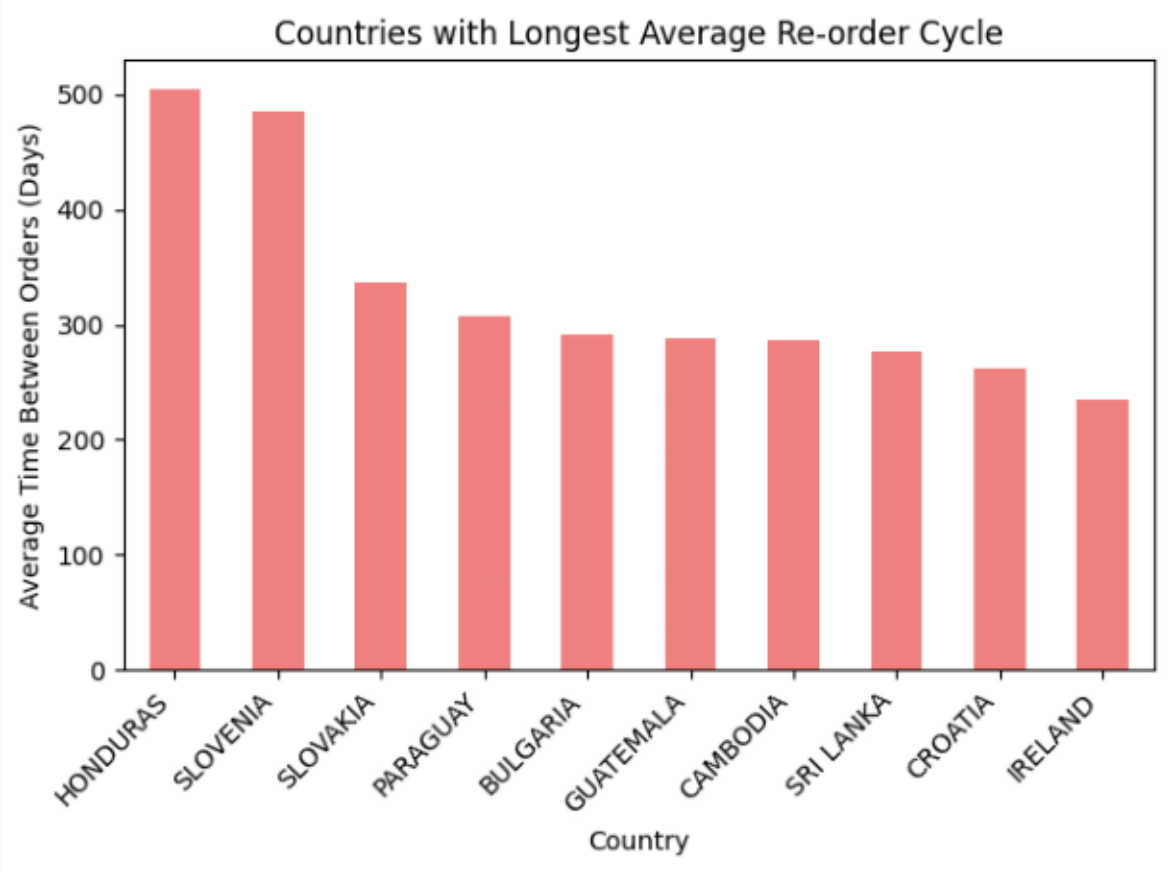
How To Explain This Graph

What this shows: Countries ranked by longest average re-order cycle (days between orders).

How to read it: Higher bars mean slower repeat demand; use this for safety-stock and replenishment strategy.

Example From Notebook Output

No textual output for this chart in notebook (visual-only output).



20) Countries with Shortest Re-order Cycle

Notebook cell: 25

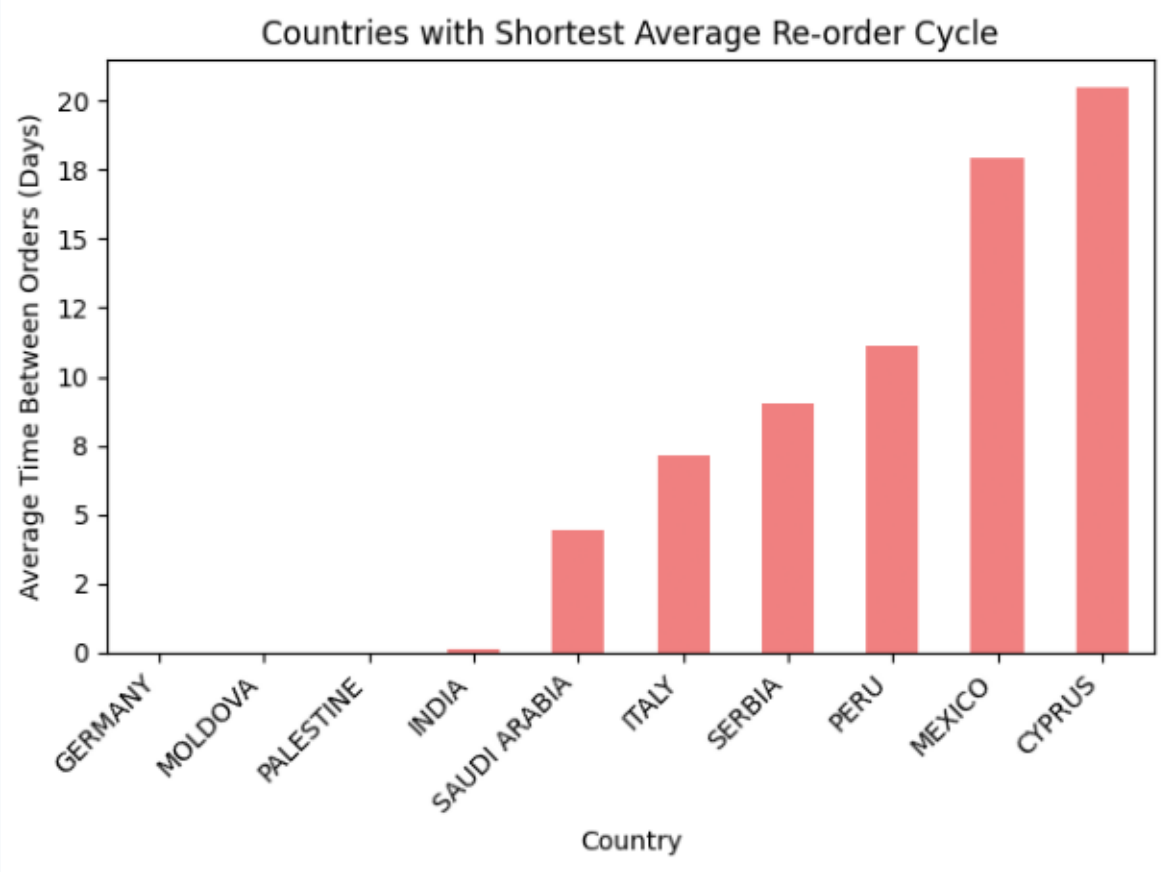
How To Explain This Graph

What this shows: Counter-view of re-order behavior: countries with shortest cycle.

How to read it: Lower cycle means faster repeat demand; these markets need tighter service and stock discipline.

Example From Notebook Output

No textual output for this chart in notebook (visual-only output).



20) Standard Compliance Cost Analysis (logistics proxy)

Notebook cell: 26

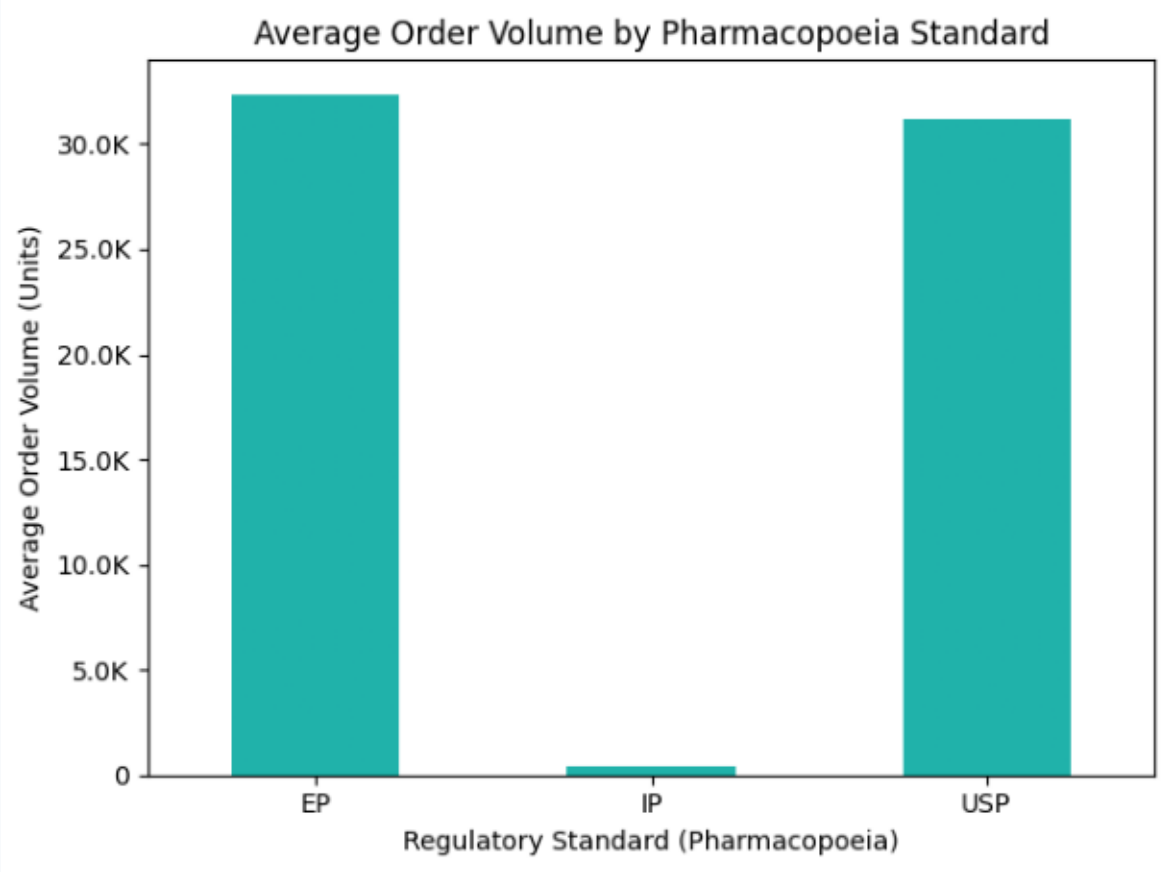
How To Explain This Graph

What this shows: Average order volume by pharmacopoeia as a logistics-cost proxy.

How to read it: Explain which standards are associated with larger batches and planning impact.

Example From Notebook Output

No textual output for this chart in notebook (visual-only output).



21) "Other" Market Investigation

Notebook cell: 27

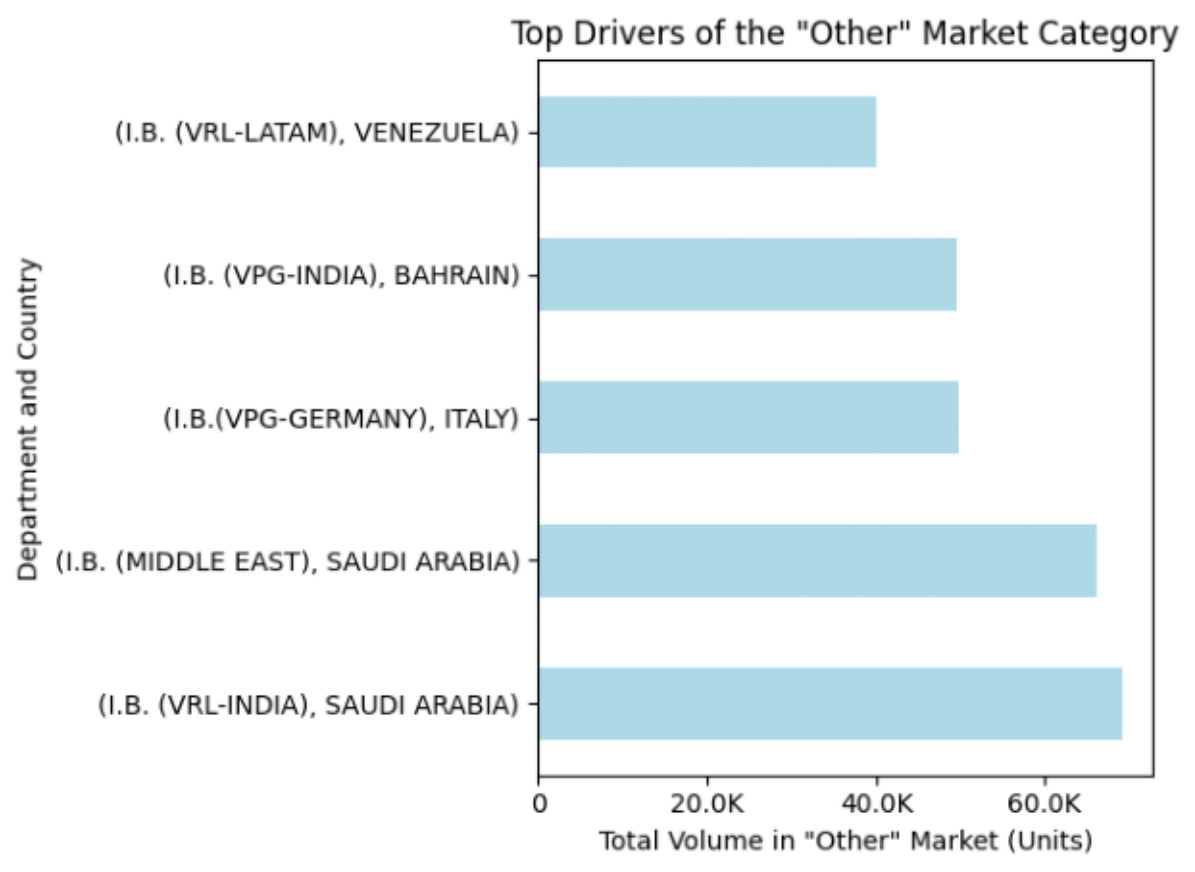
How To Explain This Graph

What this shows: Drill-down of the 'Other' market bucket by department/country.

How to read it: Use it to identify if 'Other' is meaningful business or just fragmented tail activity.

Example From Notebook Output

No textual output for this chart in notebook (visual-only output).



22) Operational Load Balance per Quarter

Notebook cell: 28

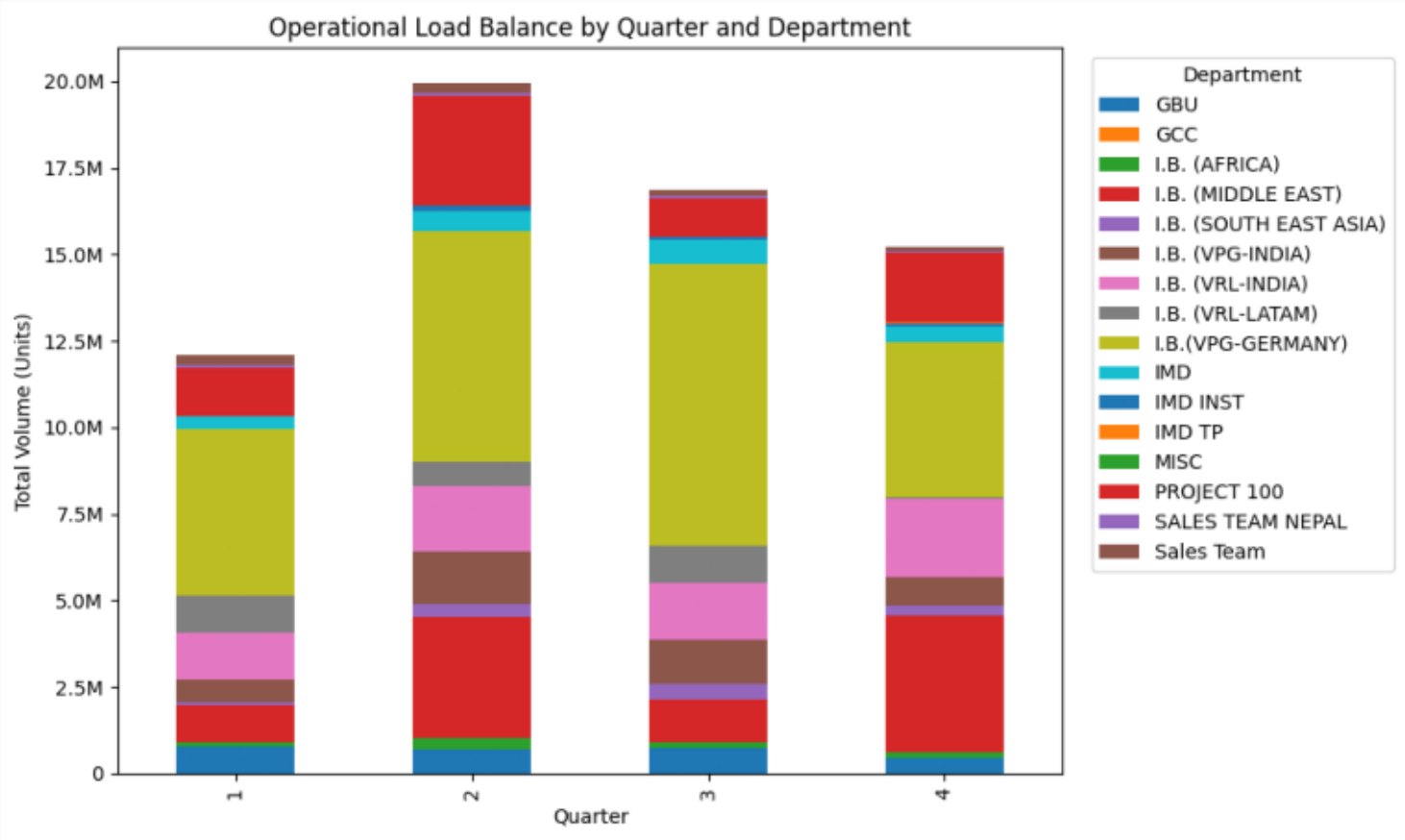
How To Explain This Graph

What this shows: Stacked quarter-wise operational load by department.

How to read it: Explain which teams carry workload each quarter and where balancing is needed.

Example From Notebook Output

No textual output for this chart in notebook (visual-only output).



23) What Is Driving the Year-to-Year Volume Change?

Notebook cell: 32

How To Explain This Graph

What this shows: Country-level contribution to latest YoY change (decliners vs growers).

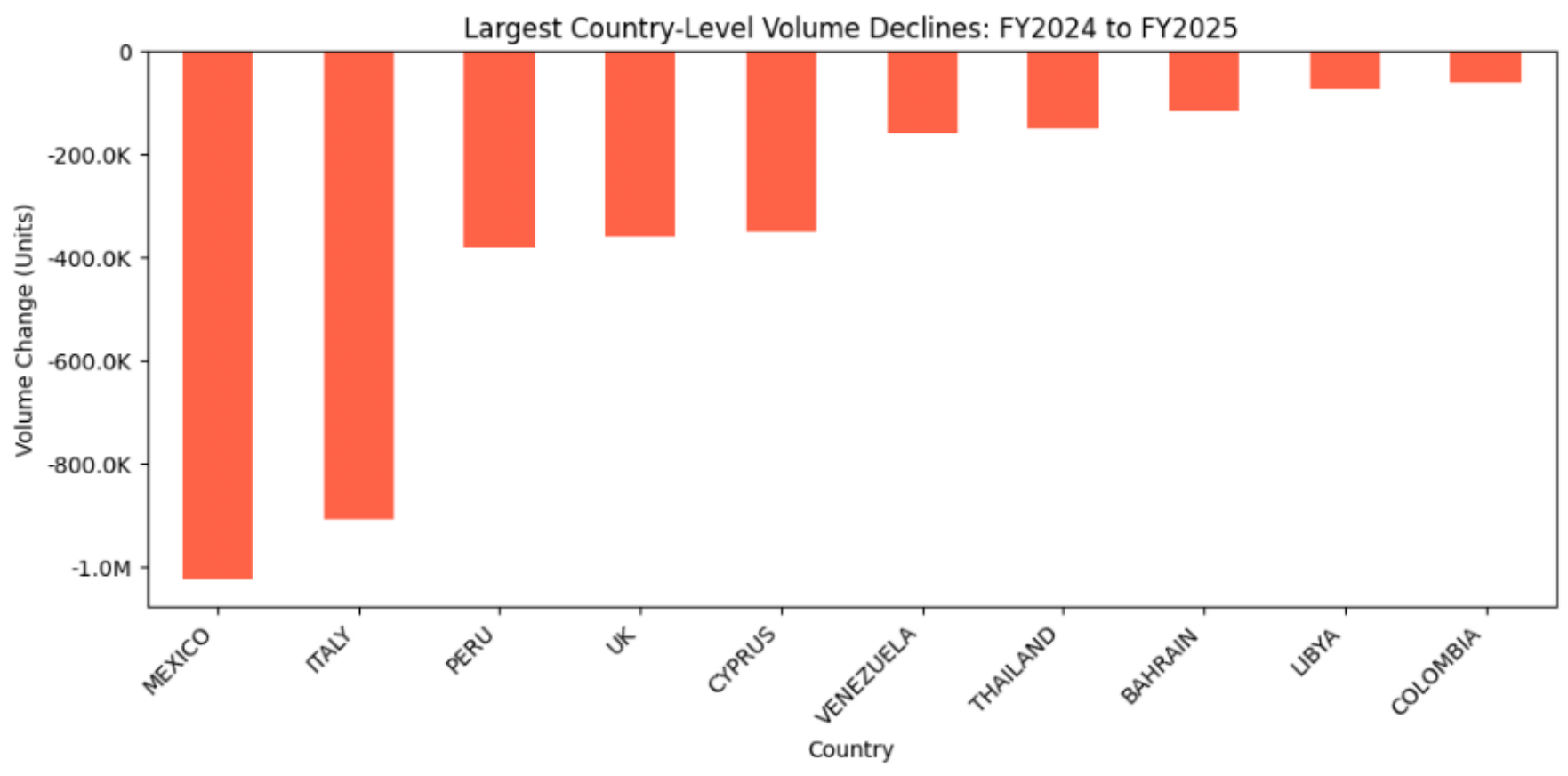
How to read it: Read negative bars as drag and positive bars as support for latest-year performance.

Business context from notebook: This shows which countries contributed most to the latest annual drop or increase.

Example From Notebook Output

Top countries reducing total volume (latest year vs previous year):

Final Country
MEXICO -1024000
ITALY -906986
PERU -381310
UK -357684
CYPRUS -349350
VENEZUELA -160000



24) How Concentrated Is the Business?

Notebook cell: 34

How To Explain This Graph

What this shows: Concentration risk trend using Top-3 share and HHI.

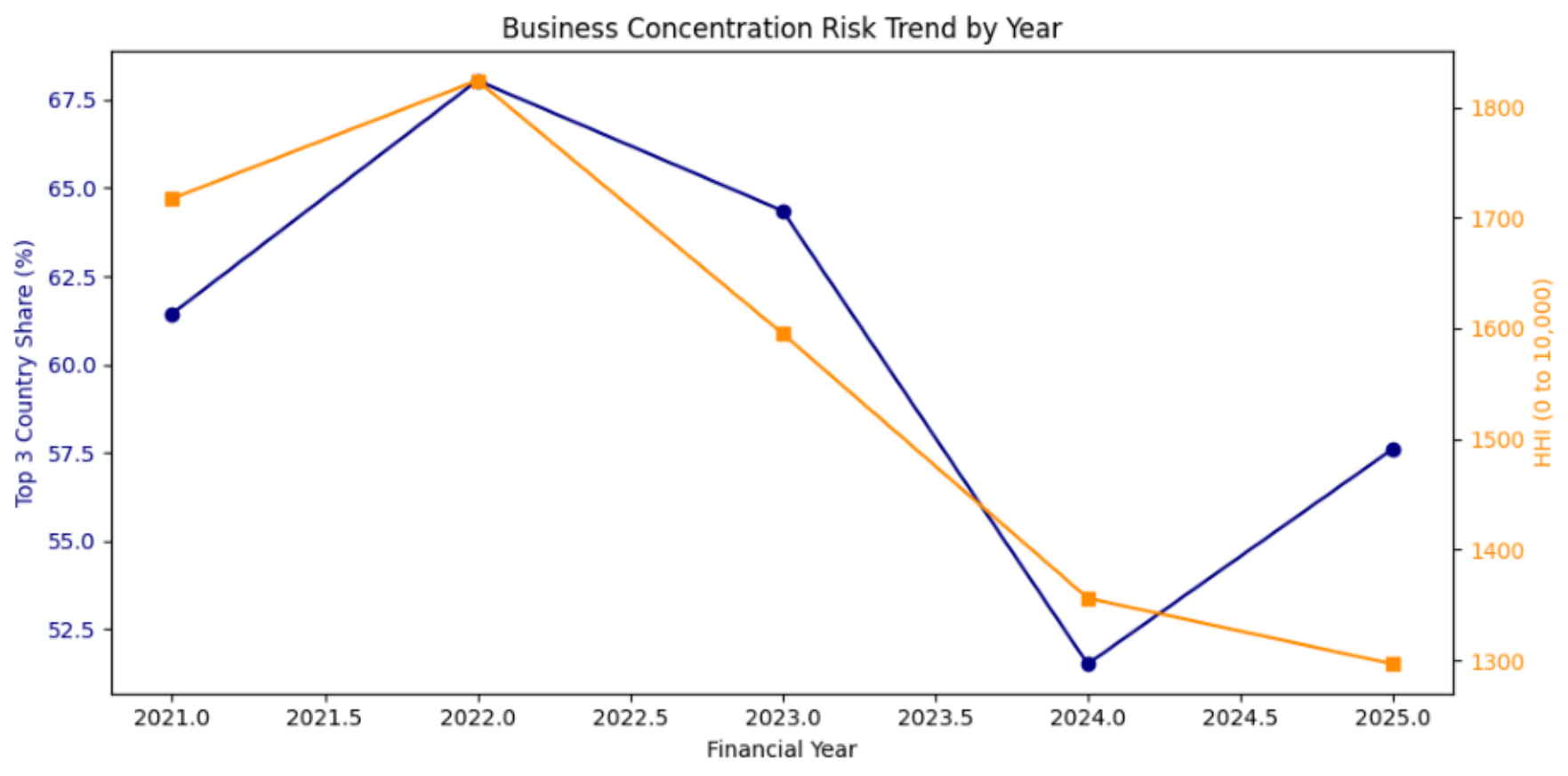
How to read it: Higher values indicate greater dependence on fewer countries; use this as risk signal.

Business context from notebook: If too much volume sits in a small number of countries, the business is more exposed to shocks.

Example From Notebook Output

Concentration trend by financial year:

Financial Year	Top 3 Country Share (%)	HHI (0 to 10,000)
2021	61.44	1717.45
2022	68.05	1824.09
2023	64.36	1595.40



25) Which Months Are Usually Strong or Weak?

Notebook cell: 36

How To Explain This Graph

What this shows: Monthly heatmap of volume by financial year.

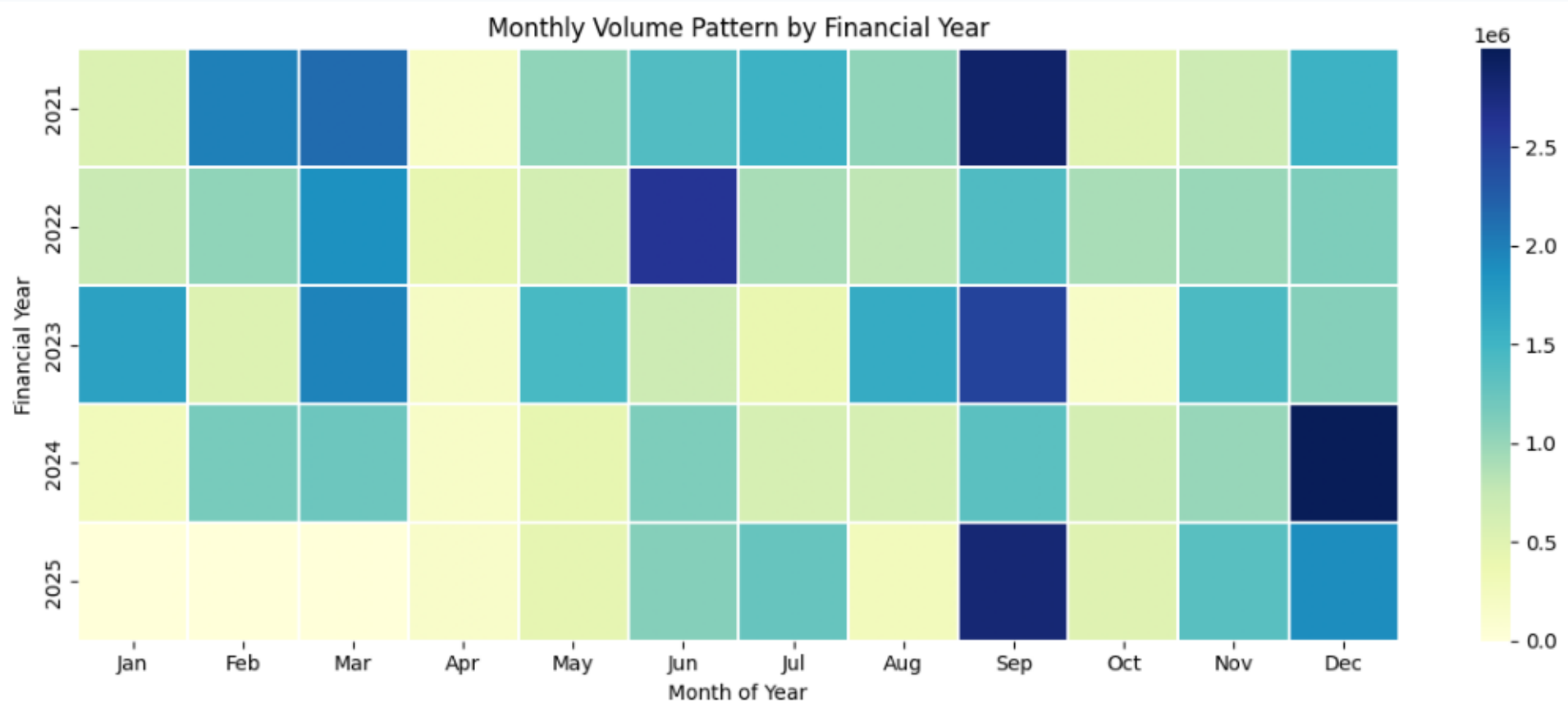
How to read it: Darker cells indicate heavier demand months; use this for production calendar planning.

Business context from notebook: A monthly heatmap helps planning teams prepare inventory and operations before peak periods.

Example From Notebook Output

Top 3 months by total volume across all years:

```
Month Name
Sep    10941128
Dec     8657688
Mar     7221491
Name: Total Qty, dtype: int64
```



26) Is Channel Mix Shifting Between Tender, Private, and Other?

Notebook cell: 38

How To Explain This Graph

What this shows: Market-mix share trend across years.

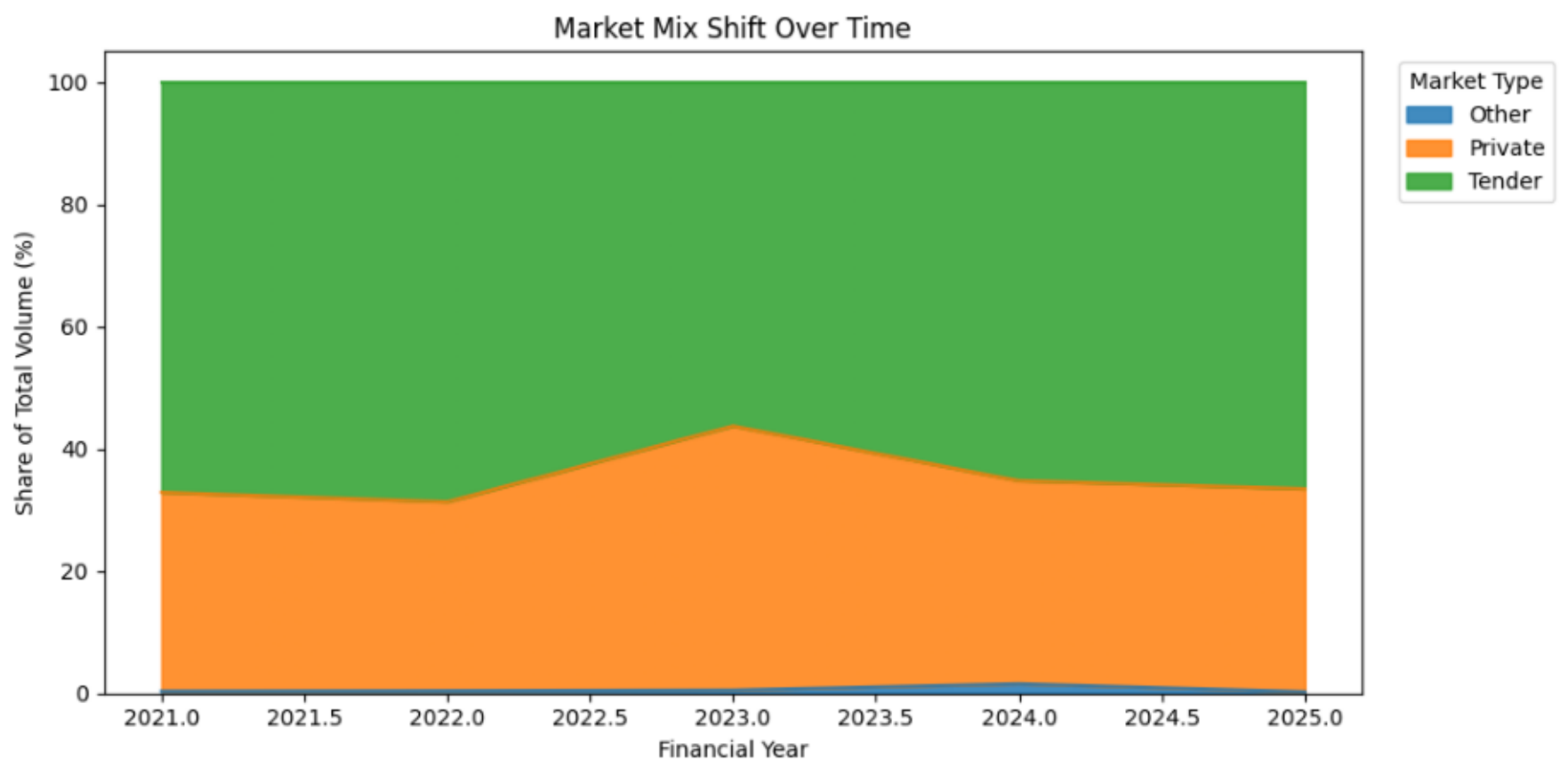
How to read it: Area movement shows channel model shift over time (Tender/Private/Other).

Business context from notebook: This shows whether the business model is changing over time.

Example From Notebook Output

Market mix by year (share %):

Market Financial Year	Other	Private	Tender
2021	0.33	32.55	67.12
2022	0.40	30.92	68.68
2023	0.50	43.23	56.27
2024	1.52	33.25	65.23
2025	0.24	33.20	66.56



27) Are New Markets Actually Sticking?

Notebook cell: 40

How To Explain This Graph

What this shows: New-market stickiness: repeat order rate after first entry.

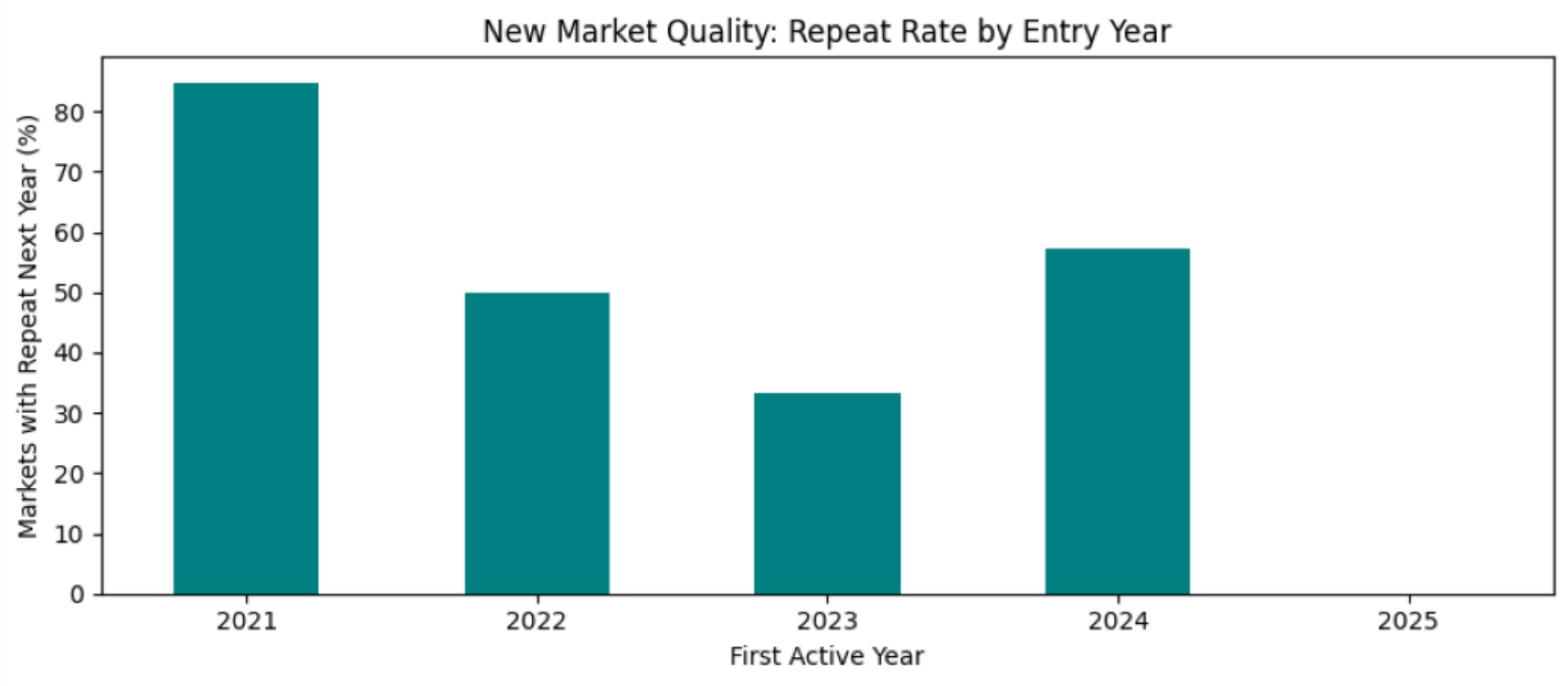
How to read it: Explain market quality: entries that repeat are scalable, non-repeat entries need review.

Business context from notebook: A market is more valuable when it places follow-up orders after first entry.

Example From Notebook Output

Recent new-market entries (last 2 financial years):

	Country	First Active Year	Repeat in Next Year
3	BULGARIA	2024	True
4	CAMBODIA	2024	True
10	EI SALVADOR	2024	False



28) Which Countries Are Stable vs Volatile in Re-order Pattern?

Notebook cell: 42

How To Explain This Graph

What this shows: Country demand stability segmentation based on re-order pattern.

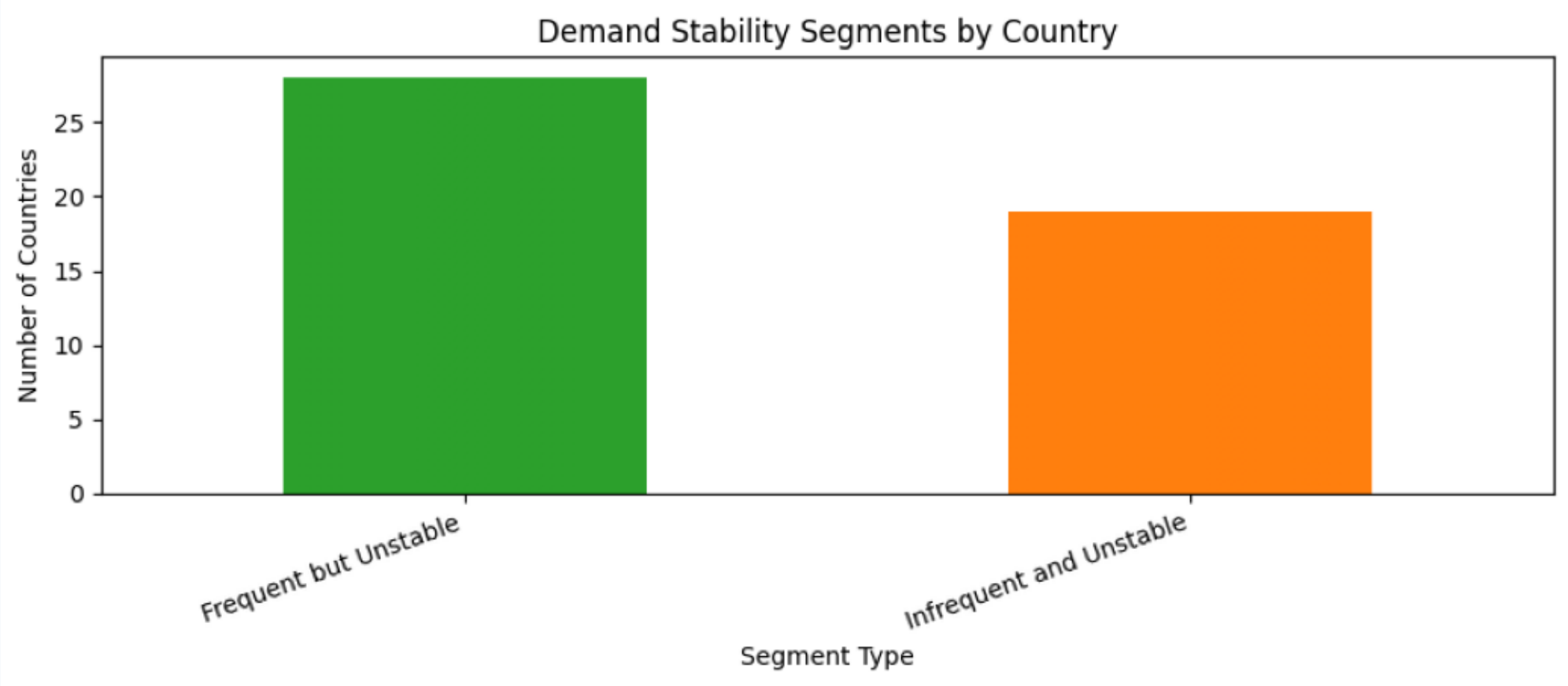
How to read it: Use segments to separate predictable markets from volatile ones for planning discipline.

Business context from notebook: This helps separate dependable markets from unpredictable ones.

Example From Notebook Output

Country stability segment counts:

```
Business Segment
Frequent but Unstable      28
Infrequent and Unstable    19
Name: count, dtype: int64
```



29) How Dependent Are We on March 31 Year-End Pushes?

Notebook cell: 44

How To Explain This Graph

What this shows: Year-end dependency: share of annual volume shipped on 31-Mar.

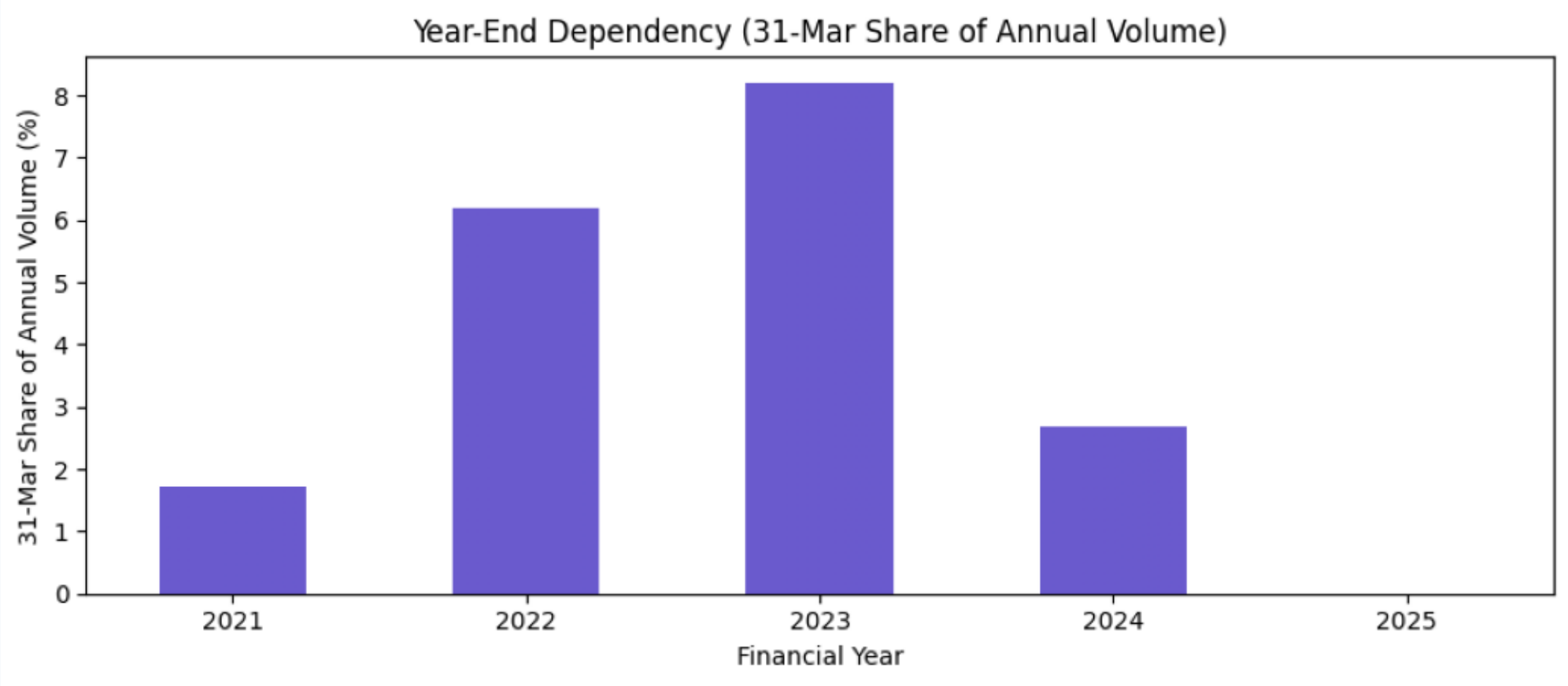
How to read it: Higher share means last-day concentration risk; lower share means smoother operations.

Business context from notebook: This quantifies how much of annual volume is concentrated on the financial year closing day.

Example From Notebook Output

Share of yearly volume shipped on 31-Mar:

```
Financial Year
2021    1.72
2022    6.20
2023    8.21
2024    2.70
2025    0.00
Name: Total Qty, dtype: float64
```



30) Is API Consumption Efficiency Staying Stable?

Notebook cell: 46

How To Explain This Graph

What this shows: Monthly API utilization ratio vs expected benchmark.

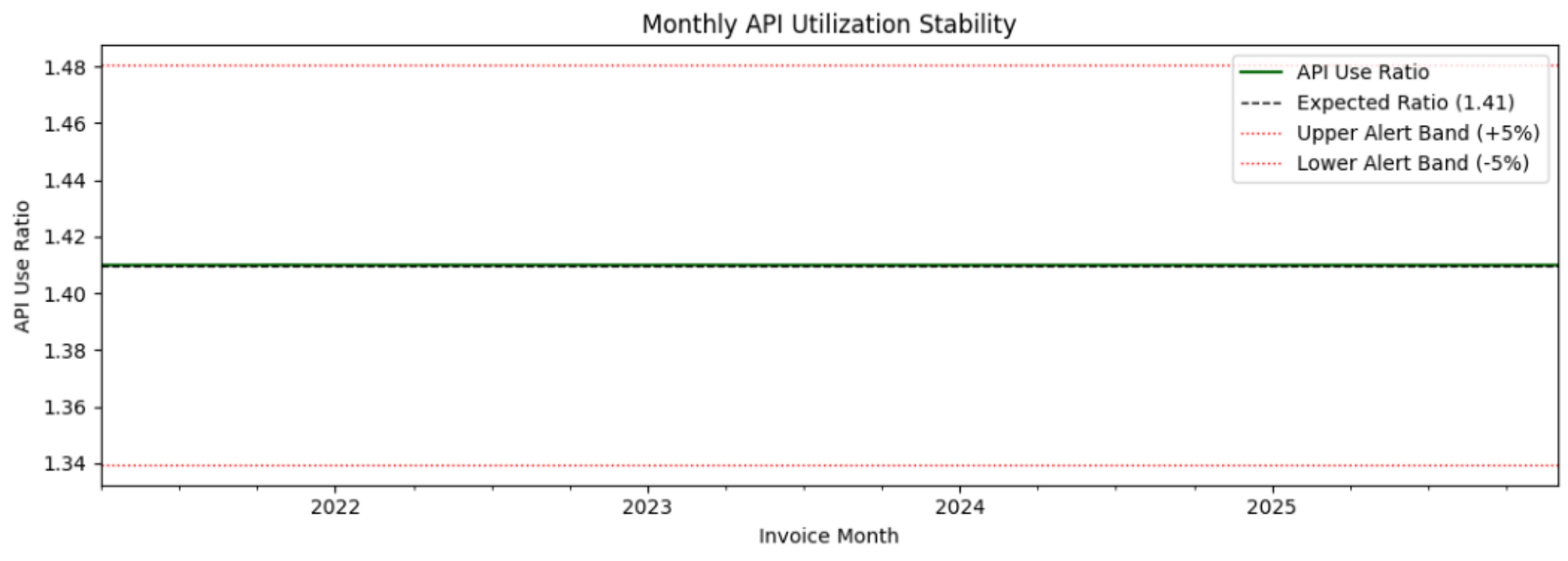
How to read it: Values near benchmark indicate process stability; sustained deviation suggests control issue.

Business context from notebook: Tracks monthly API usage ratio against the expected benchmark (around 1.41).

Example From Notebook Output

Recent monthly API utilization ratio (last 6 months available):

Invoice Date	
2025-07-31	1.41
2025-08-31	1.41
2025-09-30	1.41
2025-10-31	1.41
2025-11-30	1.41
2025-12-31	1.41



31) What Is Behind the Largest Transactions?

Notebook cell: 48

How To Explain This Graph

What this shows: Top 1% transaction outlier composition by department and market.

How to read it: Shows where extreme order concentration exists; useful for risk and capacity planning.

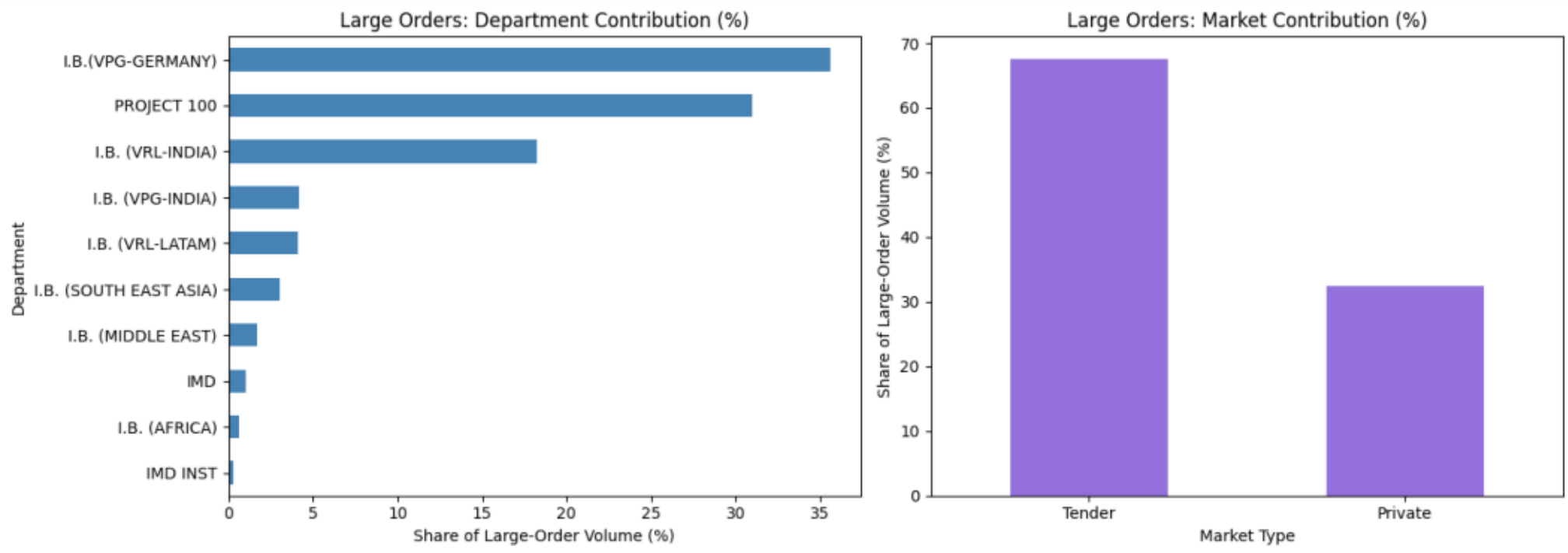
Business context from notebook: Outlier analysis helps identify whether large orders are concentrated in specific departments or channels.

Example From Notebook Output

Outlier cutoff used (99th percentile): 50,946 units

Department share inside large transactions (%):

Department	
I.B.(VPG-GERMANY)	35.61
PROJECT 100	30.97
I.B. (VRL-INDIA)	18.23
I.B. (VPG-INDIA)	4.16
I.B. (VRL-LATAM)	4.11



32) Country Opportunity Matrix: Where to Defend vs Grow?

Notebook cell: 50

How To Explain This Graph

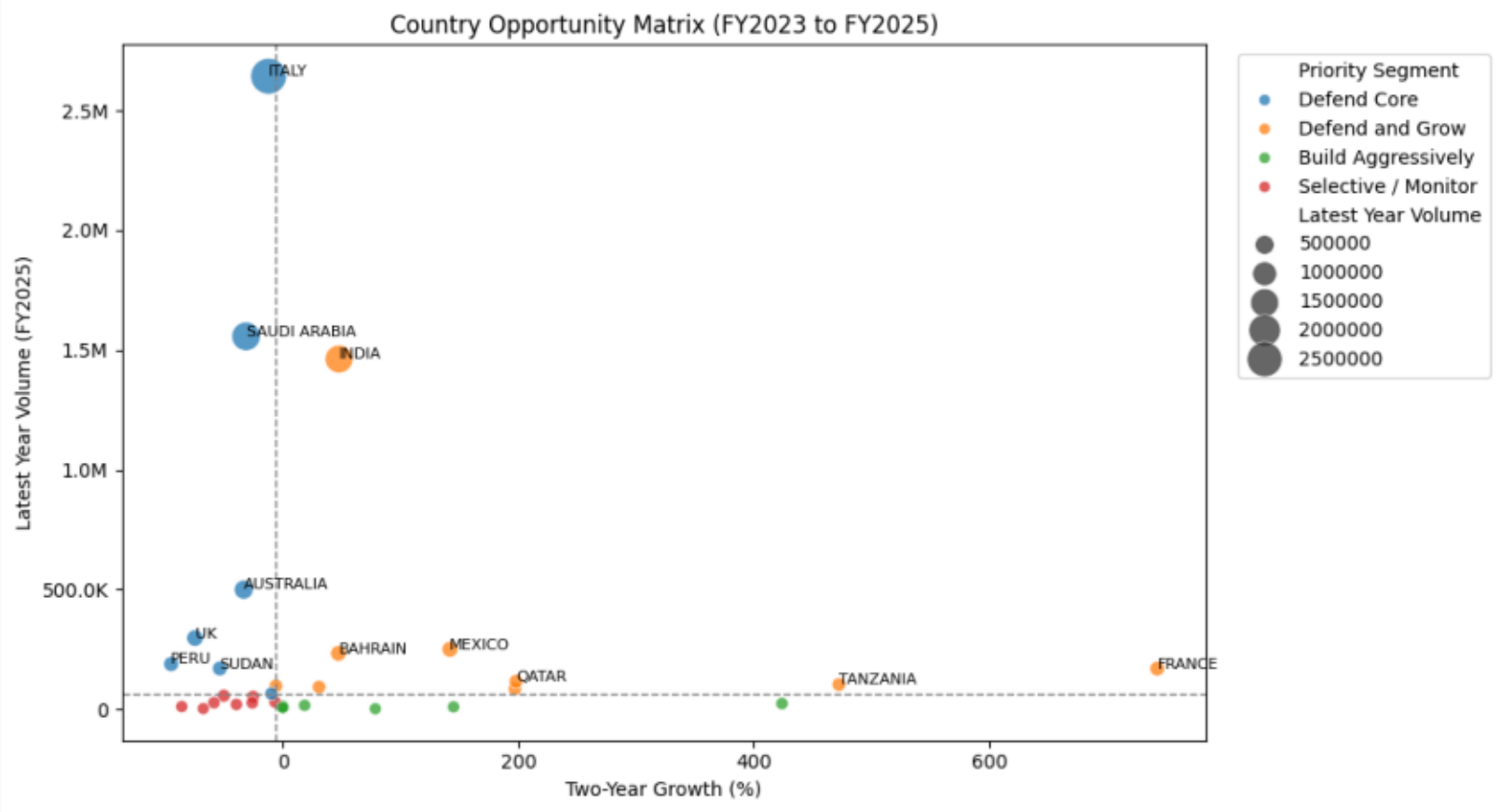
What this shows: Country opportunity matrix: latest scale vs 2-year growth segmentation.

How to read it: Use quadrants to prioritize defend-core, defend-grow, build, or monitor actions.

Business context from notebook: Combines current scale and recent growth to prioritize commercial focus.

Example From Notebook Output

Priority segment count:
Priority Segment
Defend and Grow 9
Selective / Monitor 8
Defend Core 7
Build Aggressively 7
Name: count, dtype: int64



33) Year vs Quarter Comparison - Chart 1/2

Notebook cell: 55

How To Explain This Graph

What this shows: Year-quarter comparison views (heatmap + quarter bars).

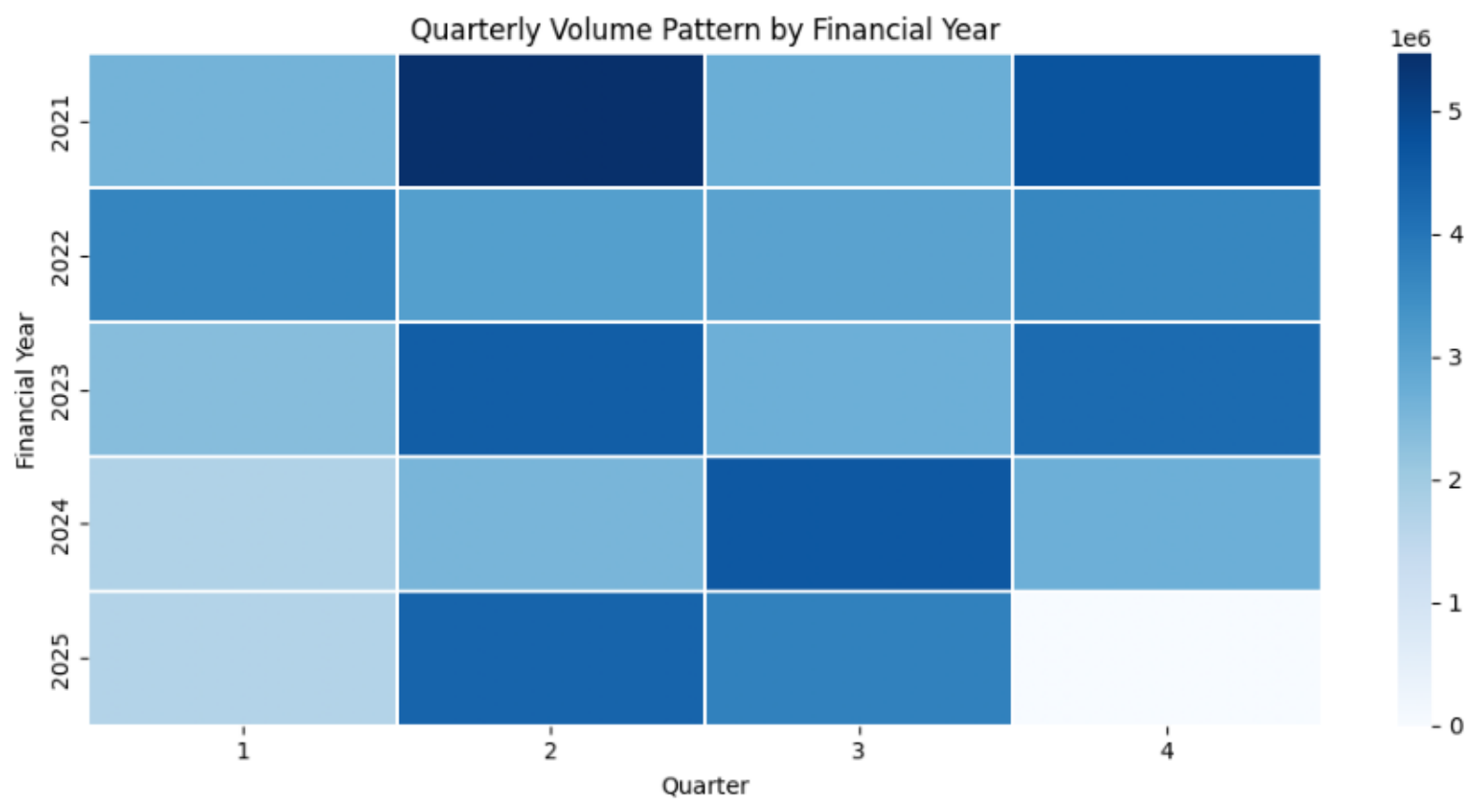
How to read it: Chart 1 shows intensity by year/quarter; Chart 2 compares quarter totals across years.

Business context from notebook: Shows how each quarter performs inside each financial year.

Example From Notebook Output

Volume by Year and Quarter (Units):

Quarter	1	2	3	4
Financial Year				
2021	2614816	5464103	2738248	4700202
2022	3686463	3106652	3025982	3631285
2023	2363110	4502637	2704779	



33) Year vs Quarter Comparison - Chart 2/2

Notebook cell: 55

How To Explain This Graph

What this shows: Year-quarter comparison views (heatmap + quarter bars).

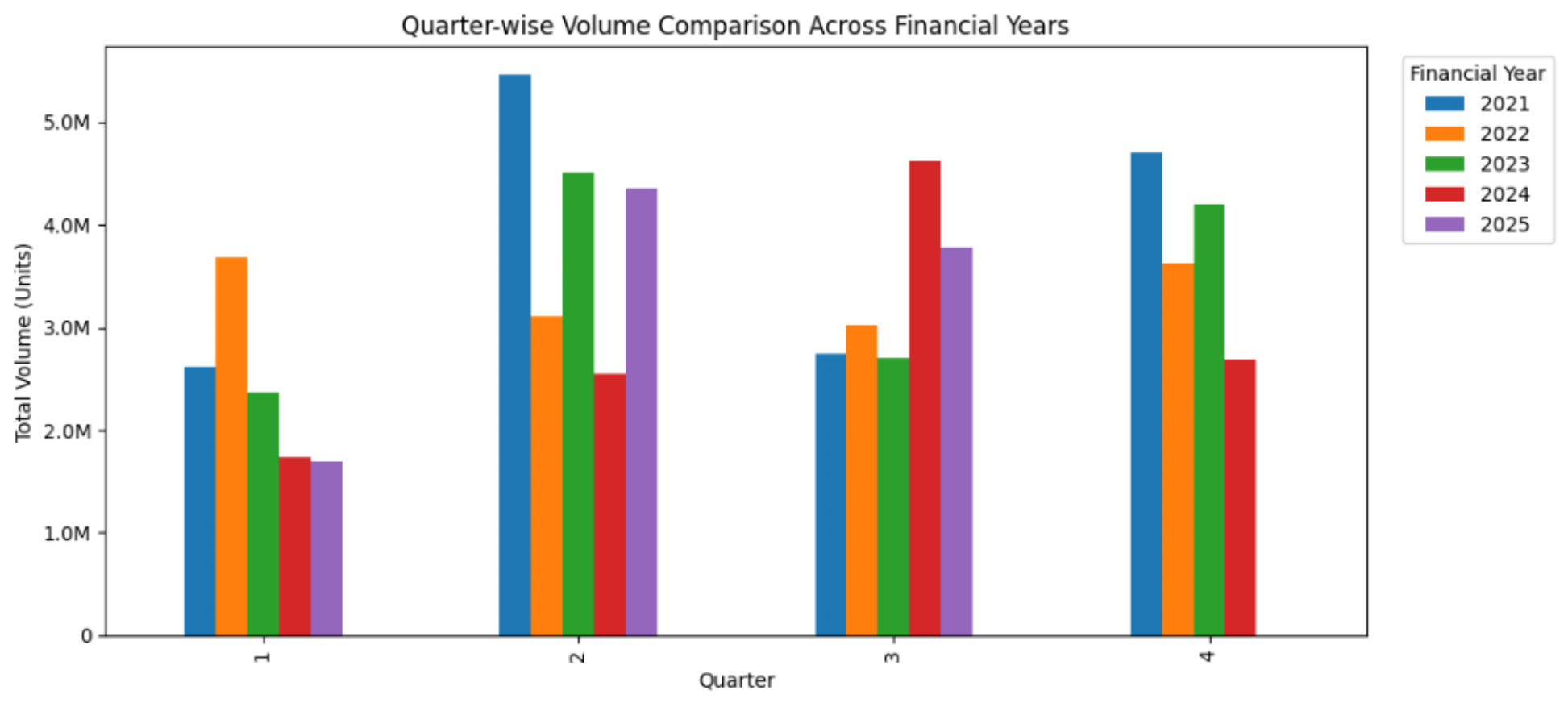
How to read it: Chart 1 shows intensity by year/quarter; Chart 2 compares quarter totals across years.

Business context from notebook: Shows how each quarter performs inside each financial year.

Example From Notebook Output

Volume by Year and Quarter (Units):

Quarter	1	2	3	4
Financial Year				
2021	2614816	5464103	2738248	4700202
2022	3686463	3106652	3025982	3631285
2023	2363110	4502637	2704779	



34) Quarter-by-Quarter Year-on-Year Change

Notebook cell: 57

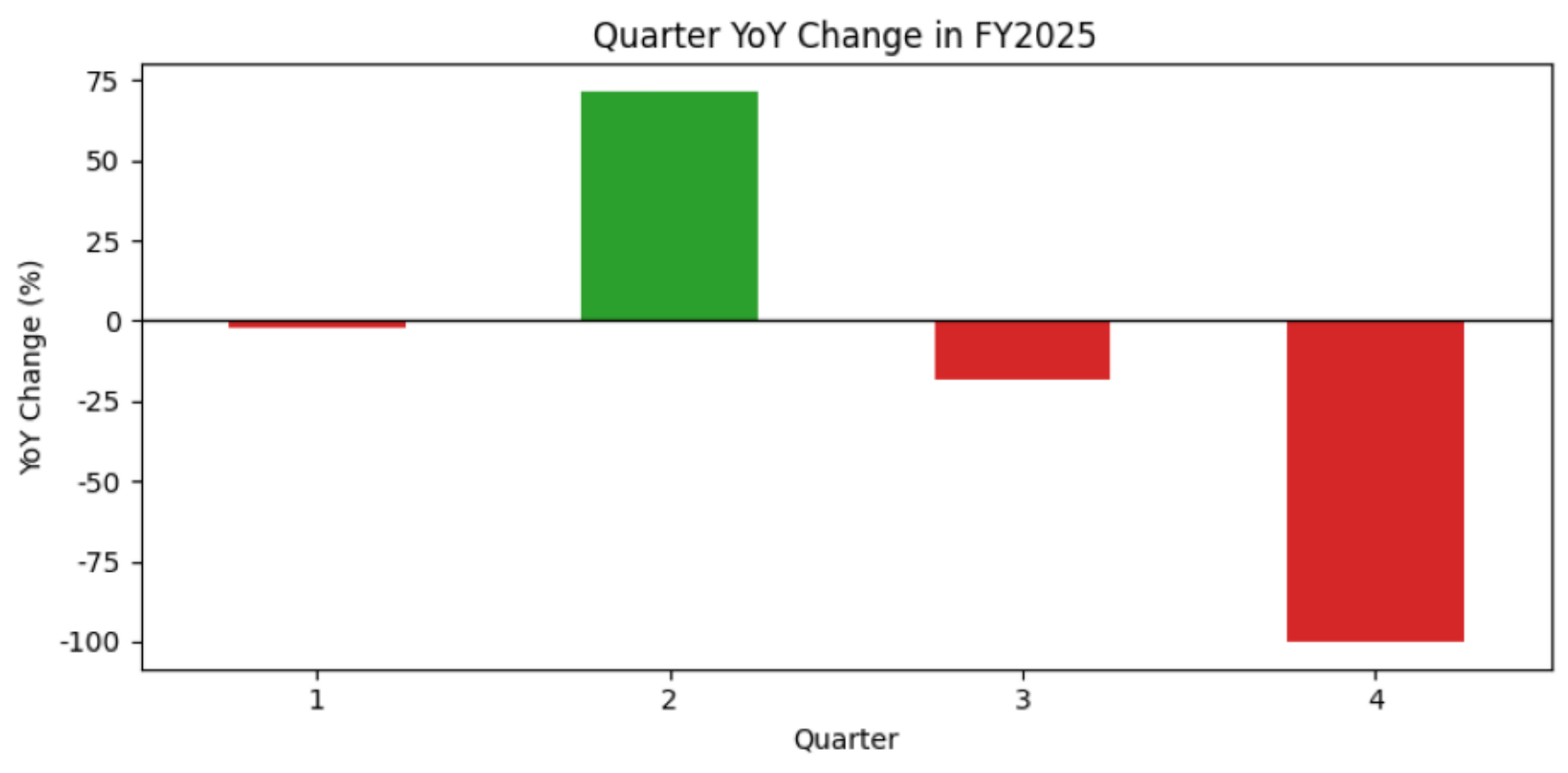
How To Explain This Graph
What this shows: Quarter-level YoY % change for latest year.

How to read it: Positive bars show improvement; negative bars flag quarter-specific correction areas.

Business context from notebook: Highlights exactly which quarter improved or dropped in the latest year.

Example From Notebook Output
Quarter-wise YoY change for FY2025 (vs FY2024):

```
Quarter
1     -2.08
2     71.39
3    -18.29
4   -100.00
Name: 2025, dtype: float64
```



35) Department Performance Across Years

Notebook cell: 59

How To Explain This Graph

What this shows: Department trend across years with growth/decline view.

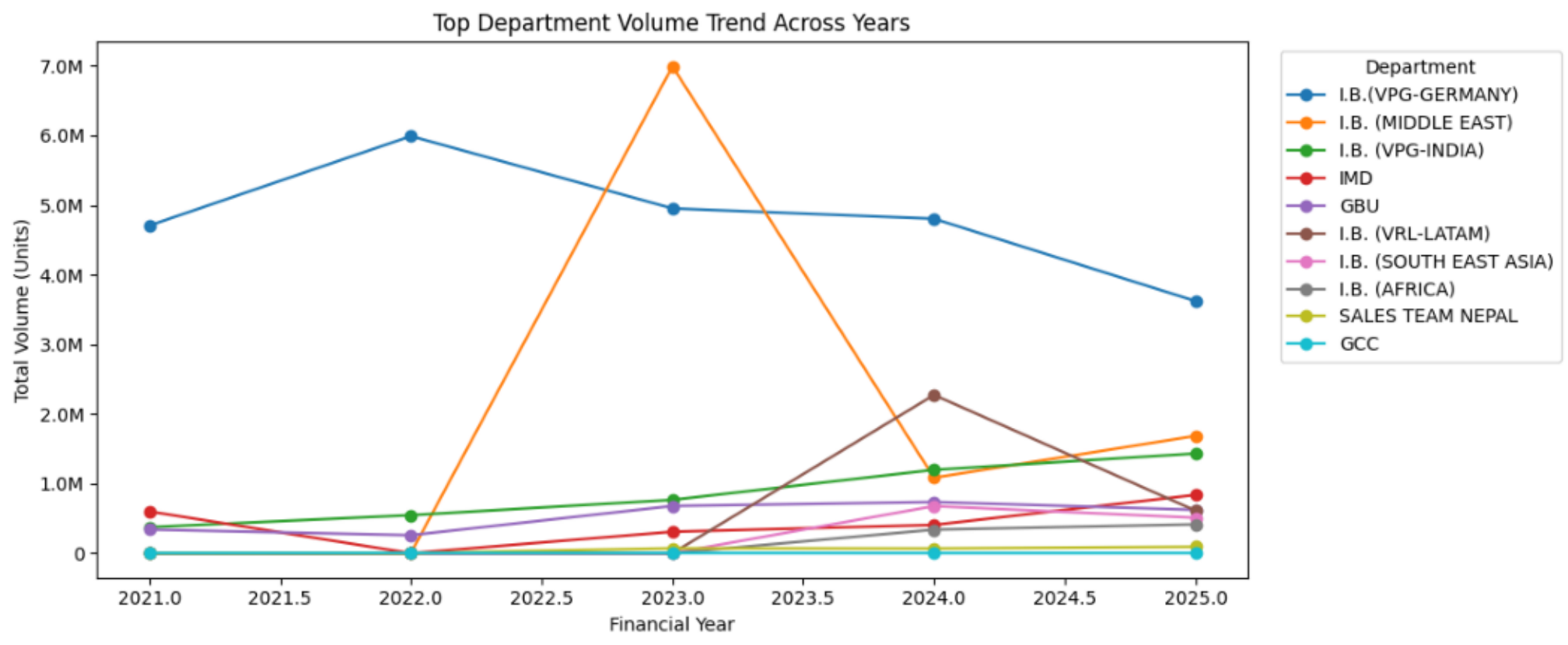
How to read it: Use this to explain structural winners and pressure points by department.

Business context from notebook: Compares every department over time and shows top gainers/decliners in the latest year.

Example From Notebook Output

Top departments based on FY2025 volume:

```
Department      I.B.(VPG-GERMANY)  I.B. (MIDDLE
EAST)  I.B. (VPG-INDIA)  \
Financial Year
2021                    4707250
0          374069
2022                    5991840
0          546954
2023                    4953536
```



36) Quarter + Department Comparison in Latest Year

Notebook cell: 61

How To Explain This Graph

What this shows: Latest-year quarter split by department (stacked).

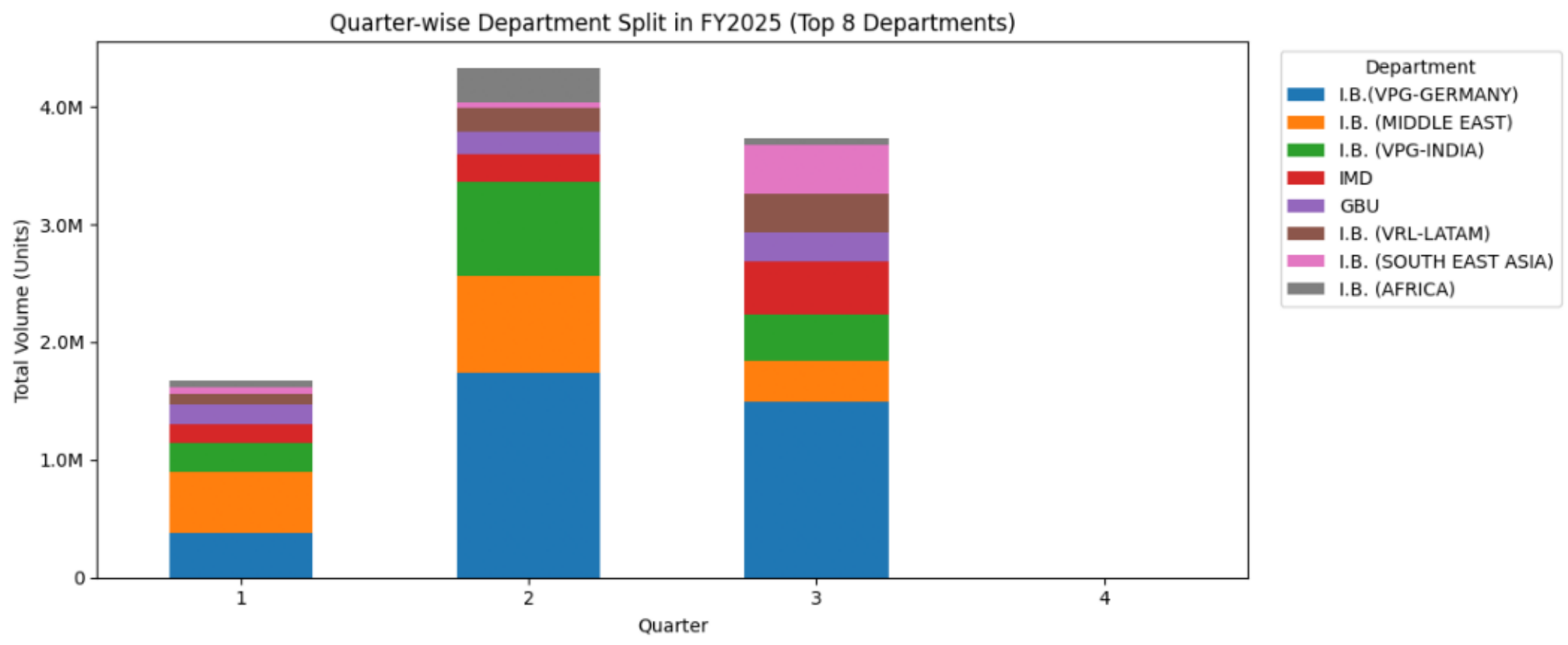
How to read it: Shows which team dominates each quarter and potential concentration in execution.

Business context from notebook: Shows which department leads each quarter in the most recent year.

Example From Notebook Output

Top department in each quarter for FY2025:

```
Quarter
1    I.B. (MIDDLE EAST)
2     I.B.(VPG-GERMANY)
3     I.B.(VPG-GERMANY)
4                   GBU
dtype: object
```



37) Strength Mix Comparison Across Years

Notebook cell: 63

How To Explain This Graph

What this shows: Strength share evolution across years.

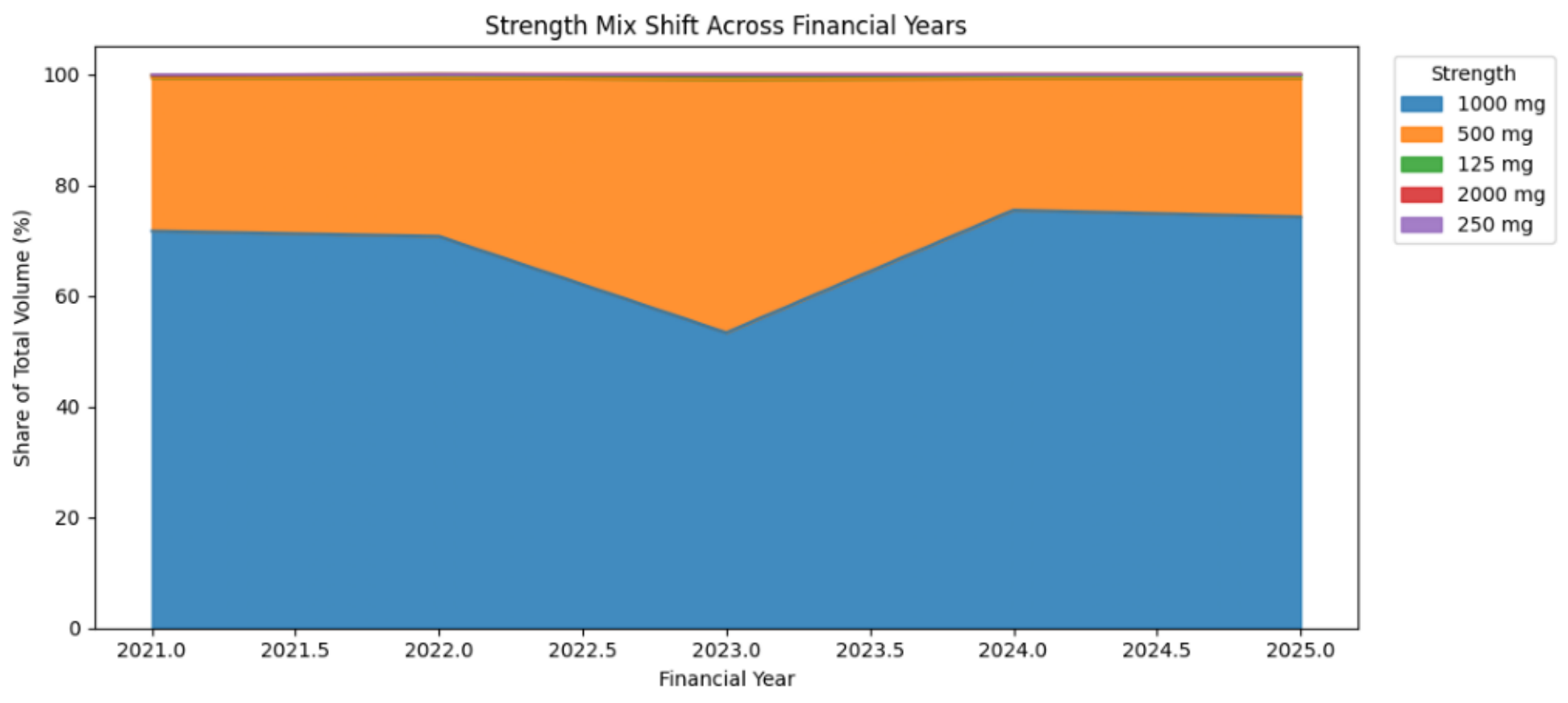
How to read it: Highlights portfolio shift between strengths and implications for manufacturing mix.

Business context from notebook: Tracks whether the business is shifting between 500 mg, 1000 mg, 2000 mg, etc.

Example From Notebook Output

Strength mix share by year (%):

Financial Year	2021	2022	2023	2024	2025
Strength Label					
1000 mg	71.69	70.71	53.33	75.46	74.22
500 mg	27.59	28.56	45.69	23.71	24.96
125 mg	0.51	0.69	0.70	0.77	



38) Department vs Strength (Latest Year)

Notebook cell: 65

How To Explain This Graph

What this shows: Department vs strength concentration (latest year heatmap).

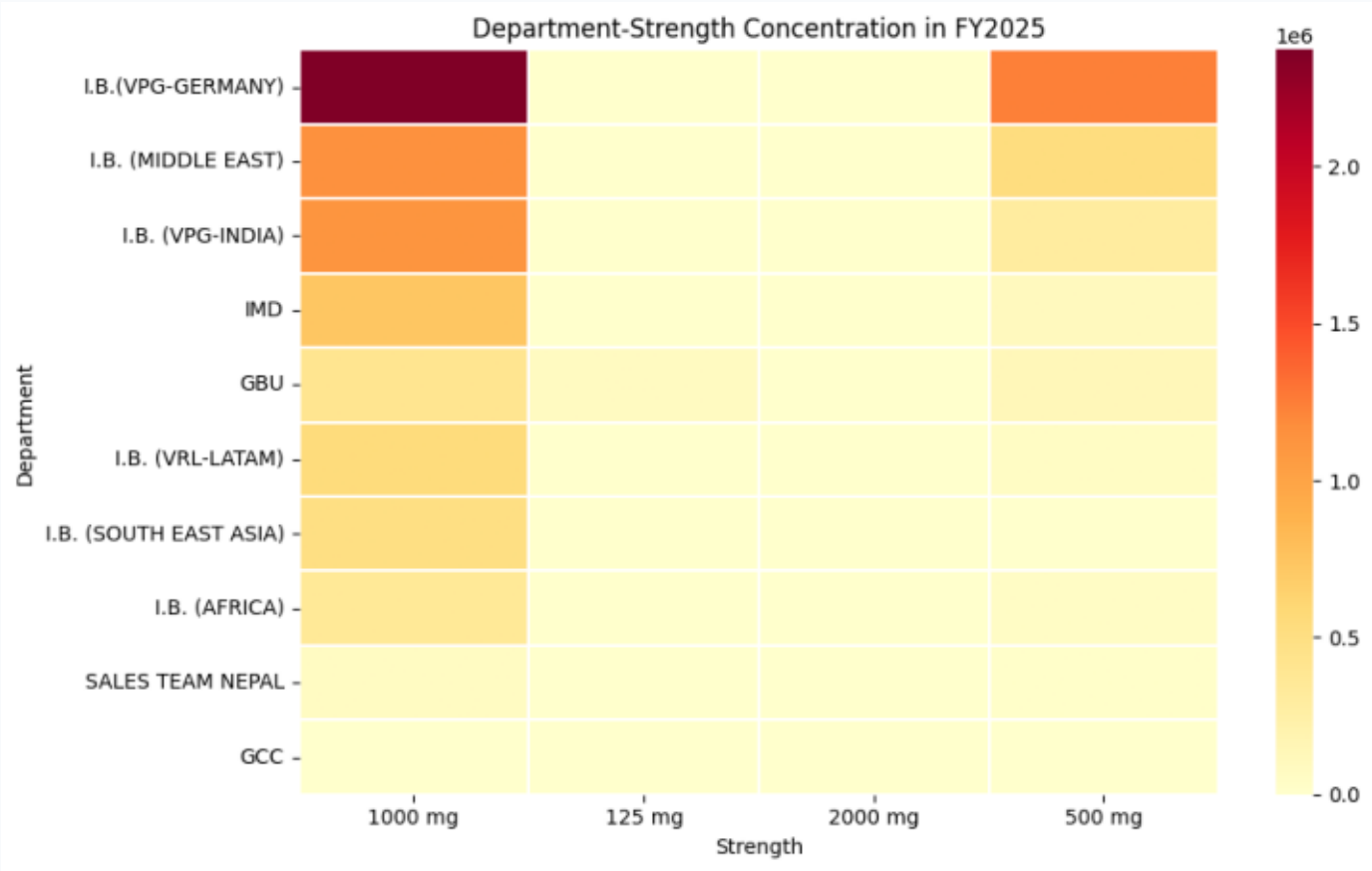
How to read it: Identify specialization: which teams are tied to specific strengths.

Business context from notebook: Identifies which departments are specialized in which strengths.

Example From Notebook Output

Department vs Strength table for FY2025 (top departments):

Strength Label	1000 mg	125 mg	2000 mg	500 mg
Department				
I.B.(VPG-GERMANY)	2372116	0	3106	1246670
I.B. (MIDDLE EAST)	1156145	0	0	529053



39) SKU Comparison Across Years (SKU = Generic Name + Strength)

Notebook cell: 67

How To Explain This Graph

What this shows: SKU trend comparison across years.

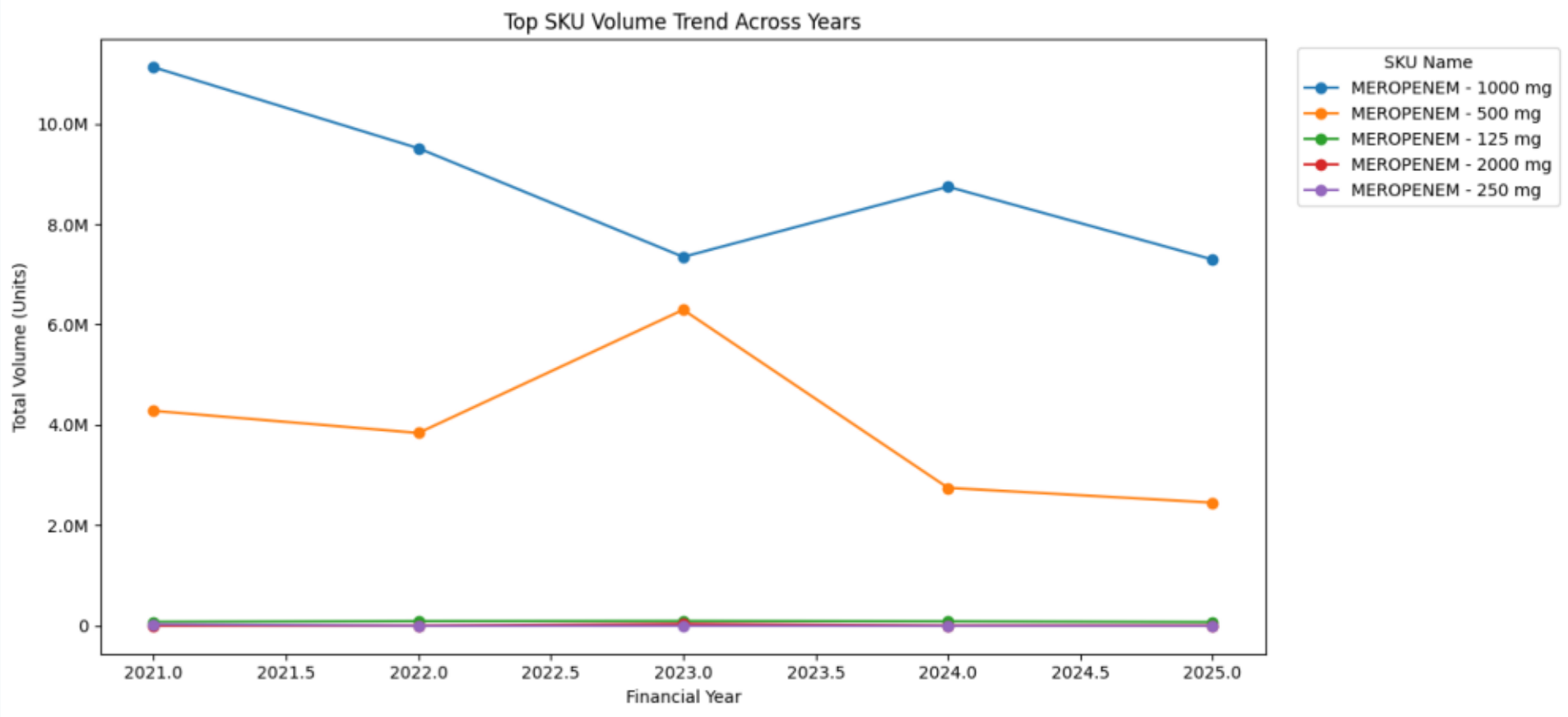
How to read it: Explain top SKU growth/decline and discuss portfolio resilience.

Business context from notebook: Shows how major SKUs are changing over time.

Example From Notebook Output

Top SKU trend table (based on FY2025):

```
SKU Name       MEROPENEM - 1000 mg  MEROPENE
500 mg  MEROPENEM - 125 mg  \
Financial Year
2021                 11124555
4281953            78797
2022                  9511246
3841284            92575
2023                  7348277
```



40) Quarter Pattern for Top SKUs in Latest Year

Notebook cell: 69

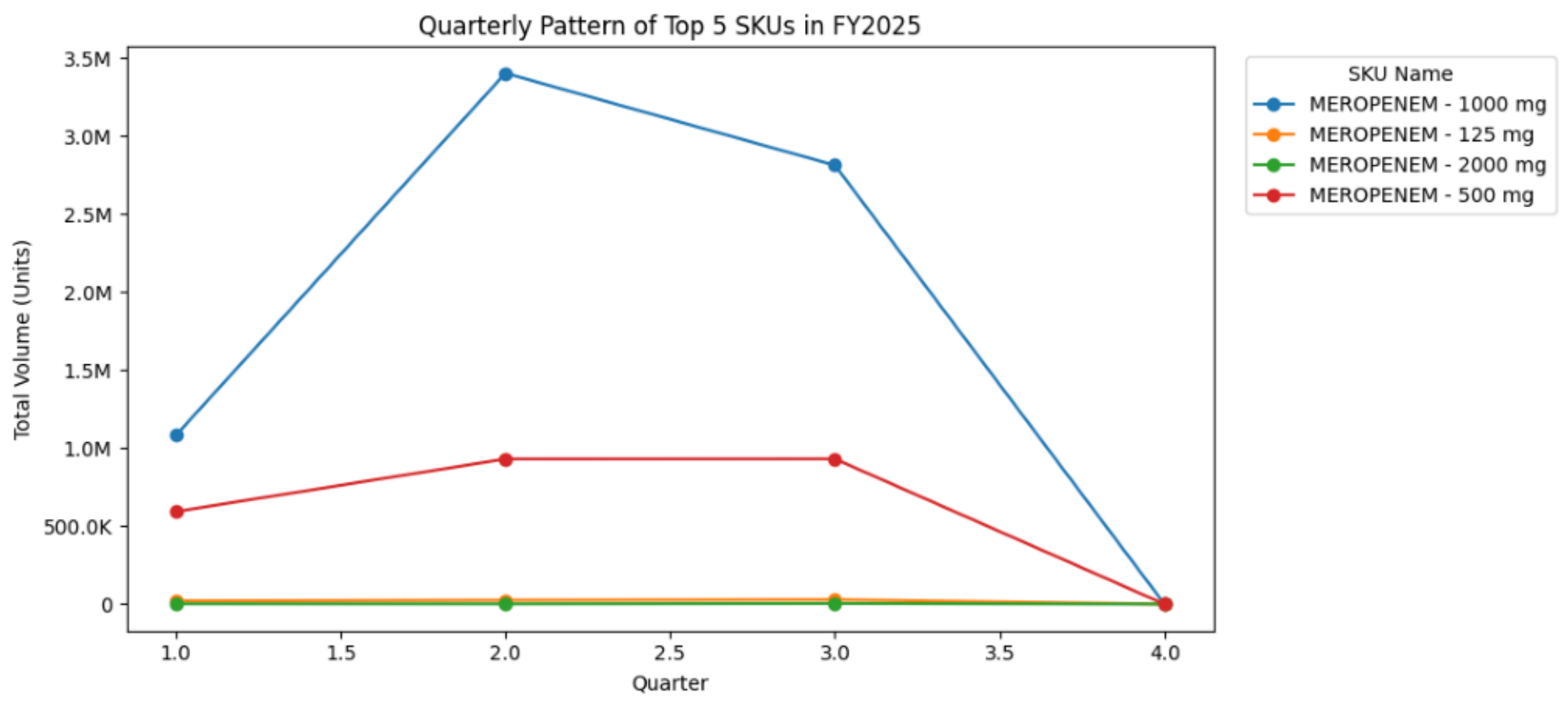
How To Explain This Graph

What this shows: Quarter pattern for top SKUs in latest year.

How to read it: Shows whether leading SKUs are steady or quarter-spike dependent.

Business context from notebook: Shows whether top SKUs are steady all year or concentrated in certain quarters.

Example From Notebook Output

Quarter pattern for top 5 SKUs in FY2025:

```
SKU Name  MEROPENEM - 1000 mg  MEROPENEM -
mg  MEROPENEM - 2000 mg  \
Quarter
1                  1081891
20700                 1620
2                  3401815
25272                  998
3                  2812081
```